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Predicting Student Dropout Risk using Machine Learning

by

Fatma Alameri

A Thesis Submitted in Partial Fulfilment of the Requirements for the Degree of
Master of Science in Professional Studies: Data Analytics

**Department of Graduate Programs & Research
Rochester Institute of Technology, Dubai**

May 2025



**Master of Science in Professional Studies:
Data Analytics**

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Abstract

Student dropout remains a persistent challenge in higher education, undermining institutional performance, reducing workforce preparedness, and limiting students' academic and economic opportunities. Accurately identifying students at risk of attrition is complex, due to the interplay of academic, financial, and behavioral factors. This thesis addresses this challenge by applying a combined machine learning framework—integrating both unsupervised and supervised techniques—to predict student dropout using structured, first-year academic and financial data.

The study utilizes a comprehensive dataset of 4,424 undergraduate student records from a European higher education institution, covering ten academic years and comprising 35 variables related to academic performance, enrollment behavior, and financial engagement. Clustering techniques were employed to group students by engagement profiles, while classification models—including Random Forest, XGBoost, and a soft voting ensemble—were trained to predict final academic outcomes: Dropout, Enrolled, or Graduate. Feature engineering was conducted in two phases, with both semester-averaged metrics and advanced derived indicators used to enhance model performance.

Findings show that academic approvals, grades, and tuition fee status are the most influential predictors of student outcomes. Unsupervised clustering revealed behaviorally distinct groups with statistically significant dropout risks, though these clusters did not translate effectively into predictive labels. Supervised models, particularly tuned XGBoost and ensemble classifiers, achieved high performance in binary classification tasks (balanced accuracy ≥ 0.91 , AUC ≥ 0.95), confirming that dropout risk can be reliably predicted from early academic records. However, multiclass classification performance declined, especially for the transitional “Enrolled” category, highlighting the limitations of static early-year data in capturing more ambiguous student states.

This research contributes to the literature by demonstrating the strengths and constraints of interpretable machine learning in modeling student success. It also offers actionable insights for academic institutions, such as prioritizing interventions for students with early signs of disengagement and financial instability. Methodologically, the study highlights opportunities for future work to explore hybrid clustering-classification models, apply soft clustering techniques, and evaluate deep learning models for benchmarking purposes. While complex models may lack interpretability, they can serve as useful baselines to understand performance ceilings

within structured educational datasets.

Keywords: Student Dropout Prediction, Feature Importance Analysis, Predictive Modeling, Clustering for Student Behavior

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Contents

Abstract	ii
List of Figures	vii
List of Tables	ix
1 Introduction	1
1.1 Background	1
1.2 Problem Statement	3
1.3 Research Aim and Objectives	3
1.4 Research Questions	4
2 Literature Review	5
2.1 Key Predictors of Student Dropout	5
2.1.1 Academic Performance Indicators	5
2.1.2 Financial and Socioeconomic Factors	5
2.1.3 Student Engagement and Behavioural Metrics	6
2.2 Machine Learning Models for Dropout Prediction	6
2.2.1 Traditional Machine Learning Models	6
2.2.2 Ensemble and Hybrid Models	7
2.2.3 Deep Learning and Neural Networks	8
2.3 Use of Unsupervised Learning in Dropout Prediction	8
2.4 Interpretability and Feature Importance in Dropout Prediction	12
2.4.1 Feature Selection Techniques	12
2.4.2 Dimensionality Reduction and Clustering	13
2.5 Temporal and Contextual Factors Affecting Dropout	13
2.5.1 Time-Based Dropout Patterns	13
2.5.2 Context-Specific Findings	14
2.6 Key Takeaways	15
3 Research Methodology	16
3.1 Dataset	16

3.2	Exploratory Data Analysis	17
3.3	Modelling	17
3.3.1	Unsupervised Learning	17
3.3.1.1	Feature Selection and Preprocessing	17
3.3.1.2	Clustering Approach	18
3.3.1.3	Outcome Validation	18
3.3.2	Supervised Learning	18
3.3.2.1	Feature Engineering	18
3.3.2.2	Modelling	20
3.3.2.3	Evaluation	21
4	Findings and Data Analysis	22
4.1	Exploratory Data Analysis	22
4.2	Unsupervised Learning	26
4.2.1	Statistical Validation of Cluster Differences and Outcomes	27
4.3	Supervised Learning	29
4.3.1	Performance Comparison Across Multi-class Classification Models . . .	29
4.3.1.1	Performance with Basic Feature Engineering	30
4.3.1.2	Performance with Advanced Feature Engineering	31
4.3.2	Performance Comparison Across Binary-class Classification Models . .	34
4.3.2.1	Performance with Basic Feature Engineering	34
4.3.2.2	Performance with Advanced Feature Engineering	36
5	Discussion	41
5.1	Comparison with existing studies	41
5.2	Research questions	42
6	Conclusion	45
6.1	Limitations	45
6.2	Future Work	46
	Bibliography	47

List of Figures

4.1	Distribution of Student Outcomes	22
4.2	Bivariate analysis of continuous features with target features	23
4.3	Bivariate analysis of categorical features with target features	24
4.4	Bivariate analysis of categorical features with target features	24
4.5	Semester progression and student outcomes	25
4.6	Multicollinearity of continuous features	25
4.7	Multicollinearity of continuous features after aggregation	25
4.8	Course-Level Difficulty Analysis	26
4.9	Cluster Projection using PCA	27
4.10	Cluster Centroids visualized using a Heatmap	27
4.11	Cluster wise influential features	28
4.12	Crosstab evaluations	29
4.13	Confusion matrix of the a Random Forest model trained on the selected features and cluster labels	29
4.14	Confusion matrix of the baseline Random Forest model using basic feature engineering for multi-class student outcome prediction	30
4.15	Confusion matrix of the tuned Random Forest model using basic feature engi- neering for multi-class student outcome prediction	31
4.16	Confusion matrix of the baseline XGBoost model using basic feature engineering for multi-class student outcome prediction	31
4.17	Confusion matrix of the tune XGBoost model using basic feature engineering for multi-class student outcome prediction	32
4.18	Confusion matrix of the ensemble model using basic feature engineering for multi-class student outcome prediction	32
4.19	Confusion matrix of the baseline Random Forest model using advanced feature engineering for multi-class student outcome prediction	33
4.20	Confusion matrix of the tuned Random Forest model using advanced feature engineering for multi-class student outcome prediction	33
4.21	Confusion matrix of the baseline XGBoost model using advanced feature engi- neering for multi-class student outcome prediction	34

4.22	Confusion matrix of the tuned XGBoost model using advanced feature engineering for multi-class student outcome prediction	34
4.23	Confusion matrix of the ensemble model using basic feature engineering for multi-class student outcome prediction	35
4.24	Multi-class model comparison	35
4.25	Confusion matrix of the baseline Random Forest model using basic feature engineering for binary class student outcome prediction	36
4.26	Confusion matrix of the tuned Random Forest model using basic feature engineering for binary class student outcome prediction	36
4.27	Confusion matrix of the baseline XGBoost model using basic feature engineering for binary class student outcome prediction	37
4.28	Confusion matrix of the tuned XGBoost model using basic feature engineering for binary class student outcome prediction	37
4.29	Confusion matrix of the tuned XGBoost model using basic feature engineering for binary class student outcome prediction	38
4.30	Confusion matrix of the baseline Random Forest model using advanced feature engineering for binary class student outcome prediction	38
4.31	Confusion matrix of the tuned Random Forest model using advanced feature engineering for binary class student outcome prediction	39
4.32	Confusion matrix of the baseline XGBoost model using advanced feature engineering for binary class student outcome prediction	39
4.33	Confusion matrix of the tuned XGBoost model using advanced feature engineering for binary class student outcome prediction	39
4.34	Confusion matrix of the tuned XGBoost model using advanced feature engineering for binary class student outcome prediction	40
4.35	Binary class model comparison	40

List of Tables

4.1	Statistical Test Results for Feature Significance	28
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Chapter 1

Introduction

1.1 Background

Higher education shapes the workforce, fosters innovation, and drives socioeconomic development. However, student dropout is a persistent challenge worldwide that affects students, academic institutions, and society. Universities are increasingly concerned with low graduation rates, as dropping out results in wasted educational resources, financial losses, and reduced student employability.

The impact of student dropout goes beyond individual setbacks. Academic institutions face challenges due to high dropout rates, negatively affecting university rankings, funding, and overall academic performance. Universities lose approximately \$16.5 billion annually in tuition revenue due to student attrition (Raisman, 2013). Similarly, state and federal governments invest heavily in education, yet significant investments in grants and subsidies are lost when students do not complete their programs.

For society, high dropout rates reduce the number of skilled professionals entering the workforce, which presents challenges to economic growth and competitiveness. In the United States, more than 65% of undergraduate students rely on student loans, and those who drop out before earning a degree are three times more likely to default on repayment McFarland et al. (2019). Furthermore, people without a degree earn 66% less than those with a bachelor's degree and are more likely to be unemployed Snyder et al. (2019). In contrast, college graduates have higher lifetime earnings, better career prospects, and improved well-being, while dropouts face higher unemployment risks and lower wages.

Despite various efforts to improve student retention, dropout rates are high. In the United States, only about 60% of full-time students graduate from their programs Snyder et al. (2019), and most students who drop out do so in their first year NSC Research Center (2019). Around 40% of U.S. students do not obtain a bachelor's degree within six years of enrollment NSC Research Center (2019). Similar trends are observed globally, with financial difficulties, academic struggles, and lack of institutional support contributing to early student attrition. In

the Sub-Saharan region, dropout rates are particularly concerning, mainly driven by financial challenges Press (2021).

Globally, student dropout is multifaceted and is influenced by academic, demographic, socioeconomic, and institutional factors. Research has highlighted several key predictors of student retention:

- **Academic Performance:** Indicators such as high school GPA, standardized test scores (SAT/ACT), early college GPA, and course completion rates are strong predictors of dropout risk. Students with lower academic success in their first semester are significantly more likely to discontinue their studies Peltier et al. (2000).
- **Demographic and Socioeconomic Factors:** Elements like age, gender, ethnicity, parental education, and financial background impact retention. Studies suggest that women are more likely to persist in their studies than men and that White and Asian students have higher retention rates compared to other ethnic groups Peltier et al. (2000). Additionally, first-generation students and those from lower-income backgrounds are at greater risk, particularly if they lack financial aid or must work extensively to cover expenses Britt et al. (2017).
- **Social Integration and Institutional Engagement:** The extent to which students are socially integrated into their university community plays a critical role. Students who build strong peer connections, participate in extracurricular activities, and receive academic support are less likely to drop out. In contrast, those who commute, live off-campus, or have limited interaction with faculty and staff face higher risks of attrition Tinto (2017).

A significant challenge in dropout prediction is that most studies focus on static data, such as transcripts and demographic profiles, or microlevel behavioral data, such as LMS engagement and attendance records. However, few studies consider meso-level factors, such as student daily experiences, social interactions, and campus engagement, which can be critical to understanding and predicting retention outcomes.

With the rise of big data in education, machine learning can enhance dropout prediction models. To be most effective, these models should integrate three key data types:

- **Academic Data:** Including GPA, attendance records, course enrollment, and standardized test scores.
- **Demographic and Socioeconomic Data:** Such as age, gender, ethnicity, parental education, and financial aid status.
- **Behavioral and Engagement Data:** Covering LMS activity, participation in student organizations, library usage, and social interactions.

By combining academic, socioeconomic, and behavioral indicators, predictive models can capture early warning signs of disengagement, allowing institutions to intervene before students officially drop out.

1.2 Problem Statement

Student retention is a challenge for educational institutions worldwide. High dropout rates impact future opportunities for students, strain institutional resources, and affect overall academic performance metrics. Institutions face difficulty identifying students at risk of dropping out, and the inability to intervene early can lead to high dropout rates among students.

Traditional methods of assessing student performance are based on historical academic records, standardized test scores, and subjective evaluations. These measures provide insight into student progress but fail to account for the broader factors that influence academic outcomes. A complex interplay of demographic, socioeconomic, and behavioral elements, such as family background, financial stability, prior academic preparation, and learning environment, shapes student success. The impact of these factors varies across individuals, making it challenging to establish a one-size-fits-all approach to student assessment and support.

Students come from different educational backgrounds, and various personal and external circumstances influence their academic trajectories. This variability makes it difficult for institutions to develop predictive frameworks that accurately assess student dropout risks and support needs.

1.3 Research Aim and Objectives

With institutions seeking effective ways to identify students at risk of dropping out and those likely to achieve academic excellence, this thesis aims to use machine learning techniques to predict student success and dropout rates based on demographic, academic, and behavioral data. By analyzing these factors, the thesis would seek to provide trends and information that can be used to support early interventions and improve student outcomes.

The primary goal of this research is dropout prediction— developing a classification model that identifies students at risk of discontinuing their studies. The model will analyze various factors to predict dropout probability. By identifying students at risk of dropping out early, institutions can design targeted support programs to enhance retention rates.

By applying clustering techniques to group students based on shared characteristics, such as attendance patterns, course performance, the thesis can categorize students into meaningful groups, such as high-risk or high-achieving. These insights can help institutions tailor strategies for different student groups.

Finally, the thesis will include an analysis of the important features to determine the strongest predictors of dropout. This will involve evaluating the impact of different features

on prediction models. Identifying key factors will allow educational institutions to focus their resources on the most critical areas affecting dropout.

This thesis aims to build a framework that minimises student dropout by integrating classification, clustering, and feature analysis. The findings of this research will contribute to decision-making in education, helping institutions develop proactive strategies to improve student retention and academic achievement.

1.4 Research Questions

1. **RQ1: What patterns in student academic and financial records are associated with different outcomes — dropout, continued enrollment, or graduation?** This question identifies trends within data that distinguish students who persist from those who disengage, highlighting the role of early academic performance, tuition status, and financial support.
2. **RQ2: Can students be grouped based on shared academic behaviors and engagement, and do these groups show different levels of dropout risk?** This question addresses whether unsupervised grouping techniques can discover categories of student behavior that correlate with varying dropout probabilities.
3. **RQ3: To what extent can student dropout be predicted using academic, financial, and enrollment-related variables available during the first year of study?** This question evaluates the predictive strength of structured student-level data in classifying educational outcomes, contributing to developing early-warning systems in higher education.
4. **RQ4: Which features most strongly contribute to the likelihood of student dropout, and how can this information help in formulating strategies for intervention?** This question aims to identify the most impactful variables influencing dropout to prioritize institutional support efforts based on evidence rather than intuition or tradition.

Chapter 2

Literature Review

2.1 Key Predictors of Student Dropout

Researchers have highlighted various academic, financial, behavioral, and socio-demographic factors as key indicators.

2.1.1 Academic Performance Indicators

Arora (2023) found that curricular units (approved and graded) were the most significant predictors. Verma et al. (2024) identified accumulated credits and failed courses as strong dropout indicators, particularly emphasizing the importance of mid-semester interventions. Alhazmi and Sheneamer (2023) demonstrated the predictive power of admission scores (GAT, AAT, high school grades) and early course performance. Matz et al. (2023) emphasized that cumulative GPA consistently emerged as a key dropout predictor across multiple institutions. Beaulac and Rosenthal (2019) identified low-grading departments, such as Mathematics, Finance, and Economics, as key predictors of student retention. Their feature importance analysis revealed that students' grades in their first five attempted credits strongly influenced their likelihood of program completion. Del Bonifro et al. (2020) also highlighted academic indicators, particularly the Additional Learning Requirements (ALR), as significant predictors. Students required to take ALR courses had higher dropout risks.

2.1.2 Financial and Socioeconomic Factors

Arora (2023) and Realinho et al. (2022) highlighted the significance of tuition fee payment status, linking financial stability to academic persistence. Dharmawan et al. (2018) found that family size and financial background influenced dropout rates, particularly in low-income households.

Across all models deployed by Hassan, Muse, and Nadarajah (2024), “grade” and “age”

emerged as the most critical predictors of dropout, with demographic features such as sex and household size also contributing significantly. In contrast, geographic features such as district and region, as well as household characteristics like housing type and occupation, demonstrated limited predictive value. Notably, some models—such as random forest and KNN—heavily relied on “grade,” raising concerns about potential overfitting and prompting the authors to apply validation techniques to ensure generalizability.

2.1.3 Student Engagement and Behavioural Metrics

Pecuchova and Drlik (2024) analyzed LMS activity logs, revealing that minimal engagement correlated with higher dropout rates. Matz et al. (2023) incorporated app-based engagement metrics, such as session frequency and event participation, to predict dropout.

Kim et al. (2023) used clustering techniques to categorize students based on dropout reasons, including employment issues, personal struggles, and academic challenges. Beaulac and Rosenthal (2019) found that engagement in structured academic programs, such as first-year seminar courses (ASSEM), significantly influenced retention rates. Students participating in these courses demonstrated higher completion rates.

2.2 Machine Learning Models for Dropout Prediction

Various machine learning techniques have been employed for student dropout prediction, ranging from traditional models to ensemble and deep learning methods.

2.2.1 Traditional Machine Learning Models

Arora (2023) tested Logistic Regression, SVM, KNN, Random Forest, and XGBoost, finding that Stacking Classifiers and Tuned XGBoost achieved the highest precision. Verma et al. (2024) used CAT, Neural Networks, and Logistic Regression, with CAT achieving an AUC of 0.853. Saa et al. (2020) found Random Forest to be the best performer, with 75.52% accuracy in predicting low-performing students. Dasi and Kanakala (2022) showed that Logistic Regression and Neural Networks outperformed Decision Trees and Naïve Bayes, achieving 93% accuracy. Del Bonifro et al. (2020) explored traditional classification models, including LDA and SVM, to predict student dropout. While these models provided reasonable accuracy when using demographic and institutional data, their predictive power was limited when academic performance metrics were not included.

Rebello Marcolino et al. (2025) methodology includes incorporating feature engineering, and multi-objective optimization using the Non-dominated Sorting Genetic Algorithm II (NSGA-II). This genetic algorithm simultaneously optimizes model selection, feature combinations, hyper-parameters, resampling strategies (notably ADASYN for class imbalance), and normalization

techniques. Several machine learning models were considered, including Random Forest, XGBoost, CatBoost, k-NN, Logistic Regression, and Naive Bayes, with CatBoost ultimately selected due to its robustness, ability to handle categorical features, and strong regularization. The use of NSGA-II facilitated the identification of Pareto-optimal configurations that balanced metrics like accuracy, F1-score, and recall, especially for the underrepresented dropout class. Five-fold cross-validation and statistical tests like the Kolmogorov–Smirnov and Mann–Whitney U were employed to ensure dataset consistency and generalizability. The results indicate that the CatBoost model, optimized via NSGA-II, significantly outperformed baseline models (e.g., XGBoost and Random Forest) across key metrics. In Experiment 2, the model achieved an F1-score of 0.86, recall of 0.85, and AUC of 0.95 for at-risk students, surpassing the earlier models and demonstrating improved reliability and predictive performance. The study showed that a single, cumulative model trained on all weeks outperformed week-specific models. SHAP analysis further confirmed the influence of features like average days with logs and maximum consecutive days without access in predicting dropout.

2.2.2 Ensemble and Hybrid Models

Kim et al. (2023) combined XGBoost with SMOTE and CatBoost with RandomOverSampler, achieving a high dropout precision of 0.963. Arora (2023) demonstrated that Stacking Classifiers outperformed Neural Networks due to ensemble methods' robustness. Alhazmi and Sheneamer (2023) found Random Forest models to be highly effective in dropout prediction, achieving high recall.

Hassan et al. (2024)'s study is grounded in empirical data from the 2022 National Education Accessibility Survey (NEAS), which collected responses from 1957 households across various regions and districts of Somaliland. The dataset includes demographic and socioeconomic variables, including household income, household size, region, district, school type, sex, age, grade level, and type of housing. The binary outcome variable defines dropout as a student who did not attain the final mark or lacked official certification for completion within a given academic year. The study used logistic regression, probit regression, naïve Bayes, decision tree, random forest, support vector machine (SVM), and K-nearest neighbours (KNN). Model training was performed using an 80/20 train-test split and 10-fold cross-validation to ensure robustness. Each model was assessed using a comprehensive set of performance metrics—accuracy, sensitivity, specificity, precision (PPV), negative predictive value (NPV), F1 score, balanced accuracy, and AUC-ROC. Hyperparameter tuning via grid and random search was used to optimise model performance, and feature importance analysis was conducted to interpret model behaviour across different algorithms. The study's results underscore the predictive efficacy of tree-based ensemble models and margin-based classifiers. Specifically, random forest achieved the highest area under the ROC curve (AUC = 0.988), signifying its superior discriminative power. At the same time, both the decision tree and SVM models posted the highest F1 scores and sensitivity

(each at 0.993), highlighting their ability to accurately classify at-risk students (Hassan et al., 2024).

2.2.3 Deep Learning and Neural Networks

Neural Networks were tested in multiple studies but often exhibited instability and inconsistencies. Arora (2023) noted that while Neural Networks achieved high recall, they lacked overall precision. Saa et al. (2020) observed that Deep Learning models performed well in classifying low performers but struggled with general classification. Kim et al. (2023) used deep learning-based clustering techniques to classify dropout reasons and tailor interventions.

2.3 Use of Unsupervised Learning in Dropout Prediction

Palani et al. (2021) present a clustering-based approach to identify low-engagement students using the Open University Learning Analytics (OULA) dataset, with the broader goal of enabling early intervention in distance learning contexts. The OULA dataset consists of over 32,000 student records from 22 courses offered between 2013 and 2014, and spans across seven interrelated CSV files containing assessments, VLE interactions, and registration data. The authors aggregated this raw data to derive three major feature categories: learning behavior (e.g., weekly click patterns across VLE tools), academic performance (e.g., average assignment scores, adjusted marks), and demographic details (e.g., age, gender). To prepare the data for clustering, the authors created derived variables such as total weekly clicks and attempted assessment weights. This aggregated data was then used to train four different unsupervised clustering models—Fuzzy C-Means, Gaussian Mixture, Hierarchical Clustering, and K-Prototype. The choice of K-Prototype was particularly well-motivated, as it is designed to handle mixed-type data, which aligns with the combined numerical and categorical nature of the dataset. Gap Statistics was used to determine the optimal number of clusters, which was found to be three. The performance of these clustering models was evaluated using multiple internal validation metrics: Silhouette Coefficient, Calinski-Harabasz Index, and Davies-Bouldin Index. Across a series of experiments, the K-Prototype model consistently outperformed the others. It achieved the highest Silhouette score (0.75), the highest Calinski-Harabasz score (17847), and the lowest Davies-Bouldin score (0.28), indicating well-defined, well-separated clusters. Interpretation of the clusters revealed three distinct student groups: Cluster 0 captured low-engagement students who showed minimal interaction with VLEs and poor academic performance; Cluster 1 included marginal students with medium engagement and low scores; and Cluster 2 comprised high-performing students with strong VLE engagement.

Iam-On and Boongoen (2017) focus on identifying clusters or profiles of students with similar academic trajectories, particularly those at risk of dropping out. The dataset for this study was extracted from MFU's Management Information System (MIS) and includes 811

undergraduate student records from the 2009 admission cohort. The data spans across four relational database views that include students' demographic profiles (e.g., sex, province, department, admission type), school-level academic performance (GPAX and subject-wise grades), and university-level academic results from the first year (GPA and grade distribution ratios across 8- and 2-level grading systems). Students were labelled either graduates or dropouts, with dropout status further detailed using institutional codes denoting academic standing (e.g., GPAX thresholds of 1.5, 1.8, and 2.0). The authors used soft subspace clustering, specifically the R-KM algorithm, to generate student groupings based on academic performance metrics. Clustering was conducted separately for two problem contexts:

1. Before university enrollment using only school-level grades
2. After the first year using both school and university performance indicators

The optimal number of clusters was determined using internal validation metrics—Davies-Bouldin (DB) and Dunn indices—over repeated trials with cluster sizes ranging from 2 to 10. Both contexts consistently indicated two optimal clusters. These clusters represented students with either high or low academic capabilities and varying dropout risk levels. In the first context, students with stronger pre-university academic profiles were more likely to graduate, with a dropout rate of just 17.05%, compared to 52.25% for those in the lower-performing cluster. In the second context, post-first-year, this differentiation became even more pronounced, with the dropout probability falling to 5.48% for high-performing students and rising sharply to 85.82% for those in the low-performing cluster. Cluster analysis showed that students with high withdrawal ratios, lower grades, and poorer entry qualifications were disproportionately represented in the high-risk group. Admission strategies also influenced dropout rates, with students admitted through regional quota (RQ) programs showing lower dropout tendencies than those admitted via direct or conditional routes.

Villar and de Andrade (2024) present a comparative study that applies supervised and unsupervised machine learning techniques to predict student academic outcomes, including dropout risk. The research builds upon a pre-existing dataset that Realinho et al. (2022) introduced, containing records of 4424 students and 35 variables. Each student's academic outcome is classified into one of three categories: Success (on-time graduation), Relative Success (graduation within three additional years), and Failure (no degree or completion beyond three years), thereby modeling dropout as part of a broader multiclass classification task that aligns well with real-world academic progression scenarios. The dataset is notably imbalanced, with the "Failure" class representing a critical minority at 32.15%. Villar and de Andrade (2024) integrate both supervised classification and unsupervised anomaly detection. For the supervised learning component, they deploy DT, RF, SVM, and four boosting models—Gradient Boosting (GB), XGBoost, CatBoost (CB), and LightGBM (LGBM). Before model training, the class imbalance problem is addressed through two synthetic oversampling methods—SMOTE and ADASYN—with SMOTE selected. Logistic regression is deliberately excluded from the modelling

pipeline to tackle the multicollinearity observed in the dataset (identified using VIF scores). Instead, tree-based and boosting models are prioritized for their performance despite feature collinearity. The authors apply the Optuna framework for performance optimisation, combining Bayesian optimisation with 10-fold cross-validation to fine-tune hyperparameters. Additionally, the study utilizes SHAP to interpret feature importance and understand model decision logic across CatBoost and LightGBM classifiers. To augment this with unsupervised learning, Isolation Forest is applied to detect anomalies, which is especially relevant for identifying hidden patterns in dropout or enrollment anomalies that supervised labels may miss. The results show the superior performance of boosting algorithms, particularly CatBoost and Lightgbm, in predicting student dropout and other academic outcomes. For the dropout class specifically, F1-scores from Random Forest, CatBoost, and Lightgbm hovered around 0.87–0.88, indicating strong model precision and recall (Villar and de Andrade, 2024). Interestingly, Optuna-based hyperparameter tuning showed limited gains in most models, although CatBoost displayed a minor F1-score improvement post-optimisation. SHAP analysis revealed consistency in top predictive features across CatBoost and Lightgbm. Meanwhile, the application of Isolation Forest for unsupervised outlier detection revealed its effectiveness in flagging dropout and enrolled students as anomalies, especially under increased contamination rates. However, the algorithm also misclassified some graduates as outliers, showing the need for refinement in unsupervised anomaly approaches when dealing with imbalanced and high-variance academic datasets.

Anwar (2025) presents a hybrid machine learning approach for predicting student dropout risk in a vocational high school setting. The study uses supervised and unsupervised techniques to identify students at high dropout risk. The dataset is derived from the Dapodik Data Centre and includes student-level information across academic, socioeconomic, and health dimensions. The features used for analysis are report card scores (S1–S3), family income, parental occupation, and distance from home to school. This dataset, covering the academic years from 2019 to 2023, was collected using random sampling and reflects the proportional representation of students who remained in school versus those who dropped out. The methodological uses K-Means clustering (unsupervised learning) and the C4.5 decision tree algorithm (supervised learning). SMOTE is used to address class imbalance due to the typically low dropout rates compared to retention. The K-Means algorithm is used first to cluster students based on relevant features—family income, distance to school, and academic grades. The optimal number of clusters is determined using the Elbow Method, which identifies where adding more clusters yields diminishing returns in within-cluster variance. Both 3- and 4-cluster configurations are tested, with the 4-cluster approach revealing greater granularity in distinguishing at-risk students, particularly in identifying a subgroup characterised by low income and declining academic performance. These clusters are then used to generate pseudo-labels for dropout risk, which are used as the target variable in a supervised learning step (Anwar, 2025). Following clustering, the study uses the C4.5 decision tree algorithm to predict whether a student is at

risk of dropping out. The input features include fixed parental income, distance to school, and subject-wise academic grades. The training and testing split is 70% to 30%, respectively. The resulting decision tree provides interpretable classification rules, with leaf nodes indicating dropout risk levels and internal nodes representing decision criteria. The results demonstrate that students with low family income, greater distance from school, and declining academic performance are likelier to drop out. The C4.5 model shows clear, hierarchical decision logic and reveals that socioeconomic and accessibility factors contribute to dropout likelihood.

N. et al. (2020) propose a predictive model integrating unsupervised clustering with supervised regression techniques to identify and forecast student dropout in Brazilian higher education institutions (HEIS). The dataset used in this study merges two national educational databases: the Higher Education Census (HEC) and the Indicators of Higher Education Flow (IHEF), both referring to undergraduate students in 2015. The HEC dataset contains detailed institutional information, while the IHEF provides statistical indicators related to student progression. The variables selected for modelling include demographic indicators, academic engagement, and financial support status (e.g., food aid, scholarship access, material assistance). The study combines K-Means clustering and regression analysis. Initially, the dataset is partitioned into clusters using K-Means, and the optimal number of clusters determined by minimising error metrics. For each cluster, a separate regression model is trained—linear regression (LR), robust regression (RR), or support vector regression (SVR)—tailored to the unique characteristics of that cluster. New instances are assigned to the nearest cluster based on Euclidean distance from cluster centroids. The combined model's prediction for a new student is generated from the regression model trained on the student's assigned cluster. This enables the model to capture intra-group variance and tailor regression predictions accordingly. Standard metrics such as Mean Squared Error (MSE) and Mean Absolute Error (MAE) are used for model evaluation across 30 randomized iterations (N. et al., 2020). The results show that the proposed model outperforms standalone regression models and their bagging ensembles, especially when using linear and robust regressors. Specifically, the PM-LR (Proposed Model with Linear Regression) achieved the lowest error values with minimal variance, confirmed by Wilcoxon signed-rank tests showing statistical significance against most competing methods. While SVR models yielded higher errors overall, they were still improved when embedded within the proposed model. The authors observe that excessive clustering (higher K values) reduces model accuracy, likely due to data sparsity within clusters. Thus, fewer, more representative clusters yielded better generalisation. Financial support indicators, such as scholarships and material aid, were identified as critical factors influencing dropout risk. Pearson correlation analysis further confirmed that dropout is negatively associated with variables indicating permanence and course completion, solidifying the connection between socioeconomic stability and academic continuity (N. et al., 2020).

Tiwari and Kumar (2022) predict student dropout in Massive Open Online Courses (MOOCs) using the publicly available KDD Cup 2015 dataset. The dataset is highly detailed and tempo-

rally rich, consisting of over 80 million log entries spanning 39 online courses and involving approximately 100,000 users. These logs capture user interactions such as video views, assessment completions, page navigations, and course closures over 30 days. A sample of 10,000 students was extracted and split into training and testing sets. The dataset was composed of several linked files—"enrollment" records, user "log" histories, "truth train" dropout labels, and metadata files about course modules and timing. Feature engineering included transformations, generation of derived variables like "course number" and "non-drop" count, and exclusion of non-contributory features like "class size" and redundant access variables (Tiwari and Kumar, 2022). Initially, hierarchical clustering segment students based on behavioural and demographic attributes extracted from the logs. This clustering step aims to reduce the variance in the data by grouping similar learners, which enhances the downstream predictive performance. The authors introduce an improved PCA (IPCA) framework for feature selection that enhances traditional principal component analysis by ranking features using eigenvalue optimisation and candidate pairwise covariance calculations. This feature-ranking step is crucial in identifying the most discriminative features influencing dropout. Following clustering and feature ranking, an ensemble classification approach is used by incorporating models like Logistic Regression (LR), Support Vector Machines (SVM), and Decision Trees (DT). A hybrid entropy measure is proposed to refine attribute selection during decision tree construction, improving interpretability and performance (Tiwari and Kumar, 2022). The experimental evaluation demonstrates that the proposed cluster-based and feature-ranked model significantly outperforms conventional methods. The Random Forest-based ensemble emerged as the best-performing model, consistently achieving accuracy values above 96% and F1-scores up to 0.98, particularly on dropout class predictions. Naïve Bayes showed comparatively weaker precision for the non-drop class. The study used 10-fold cross-validation to ensure robustness, with the confusion matrix further confirming the model's discriminative capabilities between "drop" and "non-drop" classes.

2.4 Interpretability and Feature Importance in Dropout Prediction

Understanding feature importance is crucial for making ML models interpretable and actionable in educational settings.

2.4.1 Feature Selection Techniques

Arora (2023) used the Φk correlation matrix, removing features with a correlation below 0.4. Verma et al. (2024) and Realinho et al. (2022) applied Permutation Feature Importance, emphasizing academic progress indicators. Saa et al. (2020) found that high school name was the most predictive attribute, indicating prior educational background's influence. Beaulac and Rosen-

thal (2019) used permutation-based feature importance analysis to identify key predictors of student dropout. Their findings emphasized that academic performance variables, particularly grades in specific departments, significantly influenced model predictions. Del Bonifro et al. (2020) employed Information Gain ranking to evaluate feature relevance, revealing that ALR, high school final marks, and academic school ID were among the most predictive attributes.

Goren et al. (2024) used K-means clustering to categorise university courses into four difficulty levels based on average grades and failure rates to enhance interpretability and address data sparsity in course-specific features. These difficulty clusters replaced individual course indicators with aggregated statistics per difficulty group, thereby reducing dimensionality while preserving meaningful variation. In addition, the authors introduced a “studentship” aggregation metric derived from LMS activity, split into cognitive (e.g., quiz scores, submission punctuality) and social (e.g., forum engagement, logins, video views) components. These aggregations replaced raw LMS features and provided weighted, normalised measures of student engagement behaviour.

2.4.2 Dimensionality Reduction and Clustering

Pecuchova and Drlik (2024) applied PCA and UMAP for dimensionality reduction before clustering students into engagement-based groups. Kim et al. (2023) used K-means clustering to categorize dropout reasons, enhancing model interpretability. Alhazmi and Sheneamer (2023) utilized t-SNE for performance pattern visualization, aiding in identifying at-risk students.

2.5 Temporal and Contextual Factors Affecting Dropout

Several studies examined time-based trends and the impact of contextual differences across institutions and countries.

2.5.1 Time-Based Dropout Patterns

Verma et al. (2024) found that 82% of dropout events occur at the end of semesters (July 31 or December 31), highlighting the need for early warnings. Matz et al. (2023) emphasized student persistence trends over time, showing that predictive accuracy improves with longer enrollment periods. Arora (2023) identified that Curricular Units (2nd semester approved) had the highest impact on student retention.

Goren et al. (2024) apply XGBoost and neural networks (NN) to real-world academic and LMS data from a large university over four years (2019–2022). The dataset is organised into three major data categories: Pre-Entry (demographic and socio-economic background), Learning-Behaviour (LMS-based interactions during the semester), and Achievements (final grades and academic performance metrics). The dataset also includes a novel temporal division

at five key points during the academic timeline: 4 weeks, 8 weeks, Exam A, Exam B, and the end of Semester 2, allowing the authors to evaluate predictive performance over time. Students who failed to enrol in any course for two consecutive years without completing their study track were classified as dropouts, resulting in a class imbalance. The models were trained and evaluated using an 80/20 random train-test split, and performance was assessed using AUC and dropout recall to highlight efficacy in identifying at-risk students. The results demonstrate that both models achieved high performance, particularly at later academic stages when more data, especially from final grades, were available. For example, in Semester 2, the XGBoost model achieved an AUC of 0.94 with a dropout recall of 0.76, while the NN model reached 0.92 AUC and 0.92 recall. However, early predictions (e.g., at 4 or 8 weeks) showed substantially lower performance, with dropout recall dropping to near zero for XGBoost and hovering between 0.44 and 0.72 for NN. When LMS data was excluded, pre-entry features alone were insufficient for early dropout detection ($AUC \approx 0.63$), emphasising the predictive importance of behavioural and achievement-based variables. A key insight is that academic achievement data (final grades) consistently outperformed other data categories in predictive power, although pre-entry features remained relevant until exam data became available.

Rebello Marcolino et al. (2025) predict student dropout and course failure using log data from Moodle-based Learning Management Systems (LMS). The dataset comprises 567 unique student-course instances drawn from 23 distinct courses, encompassing diverse disciplines. To ensure applicability across varying course structures, they normalized student progress as a percentage of course completion rather than relying on absolute week numbers. This dataset includes 25 features reflecting student engagement, such as log counts, access patterns, and time-to-event metrics—essential indicators of online academic behavior. Two experiments were conducted to evaluate the model's effectiveness, one with fixed 15-week durations and another incorporating dynamic, week-based features.

2.5.2 Context-Specific Findings

Verma et al. (2024) examined Finland's tuition-free education system, finding that GPA had lower predictive power compared to fee-based institutions. Dharmawan et al. (2018) analyzed non-academic factors, demonstrating the role of family background and lecturer relationships in dropout prediction. Aulck et al. (2019) studied first-year attrition trends at the University of Washington, emphasizing the importance of early academic performance. Del Bonifro et al. (2020) considered institutional factors such as high school final marks and academic school ID. These attributes indirectly reflect socioeconomic backgrounds and prior educational preparation, which contribute to dropout risks. The study found that students from lower-performing high schools faced higher dropout probabilities.

2.6 Key Takeaways

1. Key Predictors of Student Dropout Across Studies

Academic performance indicators such as cumulative GPA, failed courses, credits earned, and early-semester grades—especially in high-difficulty subjects—are the most reliable predictors across nearly all studies. Socioeconomic and financial variables (e.g., tuition status, family income) and early behavioral signals like login frequency and withdrawal rates also show strong influence, while demographic factors such as age and admission type vary in importance and are generally secondary.

2. Supervised Machine Learning Models for Dropout Prediction

Tree-based ensemble models (Random Forest, XGBoost, LightGBM, CatBoost) consistently achieve top performance with F1-scores up to 0.993 and AUCs nearing 0.99. These models benefit from robust handling of complex interactions and imbalanced data when paired with SMOTE/ADASYN, cross-validation, and advanced tuning methods like Bayesian optimization or NSGA-II. CatBoost with NSGA-II, in particular, showed exceptional generalizability across varied course structures and engagement timelines.

3. Use of Unsupervised Machine Learning in Dropout Detection

Clustering algorithms such as K-Prototype, Gaussian Mixture, R-KM, and Isolation Forest effectively uncover latent student groups based on behavior and performance, especially in the absence of labeled data. K-Prototype handles mixed data types well, yielding well-separated, interpretable clusters, while soft clustering (R-KM) and anomaly detection (Isolation Forest) expose academic risk profiles and behavioral anomalies aligned with dropout likelihood.

4. Feature Engineering and Interpretability

SHAP, Permutation Importance, and Information Gain were key tools for explaining model decisions and prioritizing actionable features. Engagement aggregations (e.g., log counts, quiz submission punctuality), difficulty encodings for courses, and dimensionality reduction (PCA, t-SNE, UMAP) were commonly applied to enhance model clarity, reduce noise, and structure inputs for both supervised and unsupervised pipelines.

5. Most Frequently Used and Effective Predictors

The most consistently impactful predictors include academic progress metrics (GPA, credits, failure history), behavioral indicators (LMS log activity, access patterns), financial variables (tuition status, family income), and selected demographic factors (age, sex, admission type). Course-level features such as assessment scores, withdrawal frequency, and difficulty clusters further strengthen model performance and interpretability.

Chapter 3

Research Methodology

3.1 Dataset

The dataset used in this thesis will be obtained from a higher education institution and will contain information about students enrolled in various undergraduate programs. It includes records from students pursuing degrees in agronomy, design, education, nursing, journalism, management, social service, and technologies. The dataset was constructed by combining multiple independent databases that capture different aspects of student information for a comprehensive analysis of factors influencing student success and dropout rates.

The data covers students enrolled between the academic years 2008–2009 and 2018–2019, following the implementation of the Bologna Process in European higher education. It comprises 4,424 student records and 35 attributes with no missing values. Each record represents an individual student and contains demographic, socioeconomic, macroeconomic, and academic data. These features provide insights into students' backgrounds, financial situations, enrollment conditions, and academic performance throughout their studies.

The dataset was compiled from several internal and external sources. Internal data were extracted from the institution's Academic Management System (AMS) and the Support System for the Teaching Activity (PAE), which track student progress and academic engagement. External data sources include the General Directorate of Higher Education (DGES), which provides information on national admission processes, and PORDATA, a database containing macroeconomic indicators such as unemployment, inflation, and GDP. These sources focus on how external factors influence student retention and success.

The target variable is the student status, which is classified into three categories: dropout, enrolled, and graduate. This classification enables the development of predictive models to help identify students at risk of dropping out and those likely to succeed academically.

3.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) will provide insights into the distribution, relationships, and patterns that may influence dropout rates. The analysis will begin with univariate exploration, examining individual attributes separately.

Visualizing categorical variables such as Gender, Nationality, Scholarship Holder, Application Mode, and Course can be done to know the frequency distributions. Binary attributes like Displaced, International, Debtor, and Tuition Fees Up to Date will provide an overview of student diversity and financial status. Visualizing continuous variables such as Age at Enrollment, Inflation Rate, and Curricular Unit Grades can help to identify their distribution, central tendency, and spread. Additionally, detecting outliers in features like GDP, Unemployment Rate, and Admission Grades will show extreme values that could influence the analysis.

Bivariate and multivariate analyses will explore relationships between variables. Correlation analysis will assess how factors like Previous Qualification (Grade) relate to Curricular Unit Grades and whether socioeconomic indicators such as Debtor Status and Scholarship Holding impact academic performance. By visualising comparisons, such as Age at Enrollment vs. Final GPA, it can be determined if older students perform differently from younger ones. These visualizations will help identify key predictors of dropout. Trend and behavior analysis will track changes over time, showing a view of student performance and dropout risks. By examining semester-wise academic progression, showing whether Curricular Unit Grades improve or decline from the first to the second semester. Comparing dropout rates between daytime and evening students will help determine whether attendance patterns influence retention. The relationship between financial support and dropout risk will show whether students with scholarships have lower dropout rates.

Finally, the analysis will examine external factors that influence student outcomes. Comparing dropout rates with macroeconomic indicators such as Unemployment Rates and Inflation Rates will reveal whether broader economic conditions correlate with student attrition, indicating the impact of external pressures on retention.

3.3 Modelling

3.3.1 Unsupervised Learning

3.3.1.1 Feature Selection and Preprocessing

Feature selection for clustering focuses exclusively on engagement-related and academic performance attributes. Demographic and socioeconomic variables are excluded to emphasize student behaviour. Key variables include attendance type (daytime or evening), curricular unit performance metrics (number of units enrolled, approved, or evaluated, credited), scholarship

status, debt and tuition fee status. These selections are grounded in prior research, notably Matz et al. (2023), which explored engagement-based clustering, and Pecuchova and Drlik (2024), who successfully categorised student behaviour using digital traces in academic systems.

Academic performance metrics were scaled using z-score normalization to ensure equal contribution to distance-based clustering algorithms, as raw values varied significantly in range and units. The original dataset already encoded categorical variables, daytime or evening attendance, and binary indicators for financial status (e.g., debtor status, scholarship holder, and tuition fee compliance) as integers. Thus, the final dataset used for clustering consisted of a combination of standardized academic features and unscaled encoded categorical indicators.

The Gower distance metric was used to accommodate the mixed data types (continuous and categorical). Unlike Euclidean or Manhattan distances, Gower distance can handle heterogeneous feature types simultaneously.

3.3.1.2 Clustering Approach

Agglomerative Hierarchical Clustering was selected as the core clustering algorithm, given its flexibility in accommodating various linkage strategies and its suitability for unsupervised discovery. The average linkage criterion was applied to build the hierarchical structure based on pairwise Gower distances. The number of clusters was determined based on the elbow scores.

The average values of each feature within a cluster were computed to profile each cluster, producing cluster centroids. A heatmap of these centroids was generated to visualize key differentiators across clusters.

ANOVA tests were conducted on continuous variables to assess the statistical significance of feature differences across clusters, and chi-square tests were applied to categorical variables. This validates whether the observed clustering patterns are statistically meaningful.

3.3.1.3 Outcome Validation

To evaluate the real-world relevance of the clusters, a crosstab analysis was conducted between the unsupervised cluster assignments and the known student outcomes (Dropout, Enrolled, Graduate). Additionally, a supervised XGBoost model was trained to predict cluster membership using the same feature set to further interpret the clustering results. SHAP values were then computed to quantify the contribution of each feature to the model's predictions.

3.3.2 Supervised Learning

3.3.2.1 Feature Engineering

Feature engineering was carried out in two tiers: basic and advanced.

The dataset contains sets of academic features, each corresponding to a different semester, and treating these as independent introduces redundancy without providing additional predictive value. Instead, combining them into semester-averaged metrics provided a temporal representation of student academic behaviour. This averaging smooths over irregularities that may arise from one-off performance fluctuations and aligns better with the objective of modelling sustained academic outcomes, such as graduation or dropout. Six composite indicators are derived through this process: average units enrolled in, credited, evaluated, and approved, as well as average grade and average units left unevaluated. Avg_enrolled is a proxy for workload or course load management, while avg_approved and avg_grade reflect achievement and academic success.

In contrast, avg_without_evaluations may indicate disengagement or administrative withdrawal from courses. By transforming raw, semester-specific features into these averages, the feature set becomes more compact and shows student progression trends. Student records that show structural inconsistencies—specifically, those labelled as "Graduate" yet demonstrating no evidence of academic success in terms of grades or approvals. The contradictions can mislead model learning.

Several columns are discarded to reduce dimensionality and mitigate the effects of multicollinearity. Original semester-level features are no longer necessary once their averaged counterparts are created. Variables exhibiting negligible correlation with the target outcome, such as nationality, international student, educational special needs, avg_credited, and avg_evaluations, are excluded to prevent them from diluting the signal of more relevant features. The same reasoning applies to macroeconomic variables, inflation, and unemployment, which are omitted at this stage, given their limited predictive contribution in preliminary analysis.

The advanced feature engineering approach includes derived features that reflect student behaviour, socio-demographic context, institutional interaction, and macroeconomic conditions. In addition to the semester-averaging process from the basic setup, aggregated indicators of credited units, enrolled units, evaluations, approvals, grades, and non-evaluated units are retained. These serve as foundational academic descriptors and are further utilized in constructing interaction-based variables. The underlying idea is to treat academic performance not as static data points, but as signals to be contextualized within a student's broader profile.

1. **Application_intensity:** Combines application mode and application order to represent how strategically or urgently a student pursued enrollment.
2. **Course_difficulty_proxy:** Multiplies the course code by average enrollment, a proxy for course workload and academic pressure.
3. **Attendance_age_factor:** Calculates the compounded influence of a student's age and attendance modality (daytime or evening), which may signal differences in work-life or external obligations.

4. **Feature_admission_performance_alignment:** Captures the consistency between a student's admission score and academic outcomes.
5. **parental_academic_weight and parental_employment_index:** The model synthesizes parental academic and employment backgrounds to interpret students' family environments as either enabling or constraining influences on academic success.
6. **Scholarship_fee_status:** It encodes scholarship holding and tuition fee compliance. This is particularly relevant in predicting student dropout, as financial stress influences attrition.
7. **unemployment_inflation, gdp_unemployment:** Captures the compound stressors that may influence student decision-making and persistence.
8. **efficiency:** The ratio of courses passed to courses enrolled in, normalized to avoid division by zero.
9. **evaluation_engagement:** The proportion of evaluated courses

Both feature engineering approaches result in an enriched feature set that combines quantitative academic indicators with proxies for engagement, alignment, socioeconomic support, and external pressures.

3.3.2.2 Modelling

Five distinct classifiers were trained and evaluated under multiclass and binary classification settings.

The dataset is split into features and target labels. The target variable is encoded into numerical classes representing the three possible academic outcomes: Dropout (0), Enrolled (1), and Graduate (2). Only Dropout and Graduate instances are retained for the binary classification task, and the target is remapped to a 0–1 scheme. In both setups, class imbalance is addressed using class weighting for tree-based models and sample weighting for boosting-based models.

The first model trained is a baseline Random Forest classifier, implemented using Scikit-learn's `RandomForestClassifier`. It is a reference point for evaluating the marginal benefits of more advanced model configurations. To improve upon the baseline, a tuned Random Forest model is developed using `RandomizedSearchCV`, which performs a randomised hyperparameter search over a predefined grid. The parameters include variations in the number of trees (`n_estimators`), maximum tree depth (`max_depth`), minimum samples per split and leaf (`min_samples_split`, `min_samples_leaf`), and the fraction of data sampled at each bootstrap step (`max_samples`). The balanced accuracy scoring metric guides this tuning process and employs 5-fold stratified cross-validation to ensure generalizability. The best-performing estimator is then selected and evaluated across folds.

Similarly, two XGBoost models are constructed, a baseline version and a tuned version. The baseline XGBClassifier is trained with `objective='multi:softprob'` for multiclass settings or `binary:logistic` for binary classification. It is configured with `enable_categorical=True` to handle encoded categorical features efficiently. Sample weights are calculated to reflect class imbalance and are passed into the model's training phase to further correct for skewed label distributions. The tuned XGBoost model is obtained through `RandomizedSearchCV` using a hyperparameter grid that includes boosting parameters such as learning rate, maximum depth, child weight, subsample ratio, column subsampling, and regularisation (`gamma`). The tuning uses stratified 5-fold cross-validation to estimate performance on unseen data and is optimised using the same `balanced_accuracy` metric.

The final model is a soft voting ensemble, combining the tuned Random Forest and XGBoost classifiers using Scikit-learn's `VotingClassifier`. This model computes class probabilities by averaging those from each constituent model and selecting the class with the highest average probability. The ensemble model uses the complementary strengths of the two classifiers: Random Forest offers greater stability and interpretability, whereas XGBoost is often more sensitive to fine-grained patterns in the data. Their combination aims to reduce model variance and improve predictive performance by aggregating diverse decision boundaries.

In the binary classification setting, all models are evaluated using stratified 5-fold cross-validation, where model performance is averaged across five folds to ensure stability. In the multiclass setting, initial model development is conducted using a single train-test split to facilitate rapid experimentation, although the evaluation strategy may be extended to cross-validation in subsequent analysis phases.

3.3.2.3 Evaluation

balanced accuracy, macro-averaged F1 score, and multiclass area under the ROC curve (AUC-OVR) are used to assess the models' effectiveness. Balanced accuracy accounts for imbalanced class distributions by averaging recall scores across all classes. The macro F1 score measures the balance between precision and recall for each class without favouring the majority class, making it especially suitable for multiclass and imbalanced scenarios. Finally, AUC-OVR evaluates a model's discriminative ability across multiple binary thresholds. All metrics were computed using stratified 5-fold cross-validation to ensure stability and generalizability of the results.

Chapter 4

Findings and Data Analysis

4.1 Exploratory Data Analysis

Figure 4.1 shows the proportion of students across the three target classes: Dropout, Enrolled, and Graduate. The chart shows a moderately imbalanced class distribution, with 50% of students classified as Graduated, 32% as Dropouts, and only 18% remaining Enrolled.

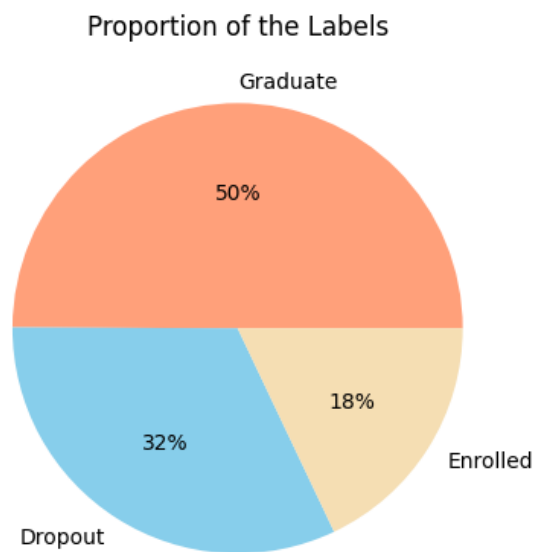


Figure 4.1: Distribution of Student Outcomes

Figure 4.2, 4.3, and 4.4, show a series of bivariate visualisations generated across all categorical and numerical variables. Key academic indicators, particularly semester-wise grades, approved units, and evaluations, exhibited a strong gradient across the three outcome classes. Graduates consistently scored higher in both first and second semester grades and approved more curricular units than dropouts, who trailed significantly in engagement and performance metrics. These trends solidify the relevance of academic performance as a critical and primary

predictive signal of at-risk students.

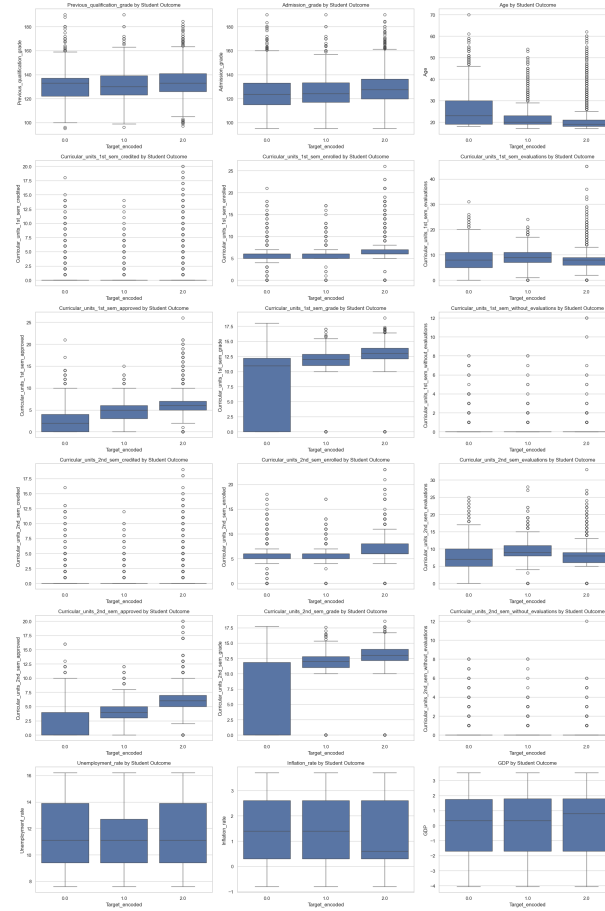


Figure 4.2: Bivariate analysis of continuous features with target features

Among categorical features, financial indicators were prominent correlates of dropout behaviour. Students flagged as debtors or with unpaid tuition fees showed higher dropout frequencies than financially compliant students. Likewise, scholarship holders—presumably those receiving financial aid—were more likely to graduate, suggesting that targeted financial support plays a mitigating role in attrition. Other notable associations include attendance mode and gender: daytime students and females showed marginally better academic outcomes.

Parental education and occupation, nationality, and application-related attributes showed wide variability with less consistent patterns, indicating weaker direct influence on immediate academic outcomes. While these variables capture relevant socioeconomic context, their noisy or high-cardinality nature suggests the need for dimensionality reduction or hierarchical grouping before integration into predictive pipelines.

Figure 4.5 shows academic performance through the semesters. For each academic metric—enrolled units, evaluations attempted, approvals, grades, and units without evaluations—students largely cluster near the diagonal reference line, suggesting strong linear concordance between Semester 1 and Semester 2 performance. Students tend not to fluctuate drastically in engagement from one semester to the next, and those who do are typically outliers. The most notable divergence appears in evaluations attempted, where several students

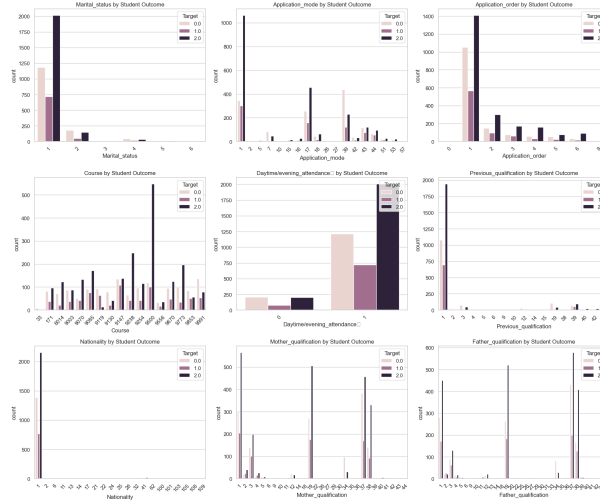


Figure 4.3: Bivariate analysis of categorical features with target features

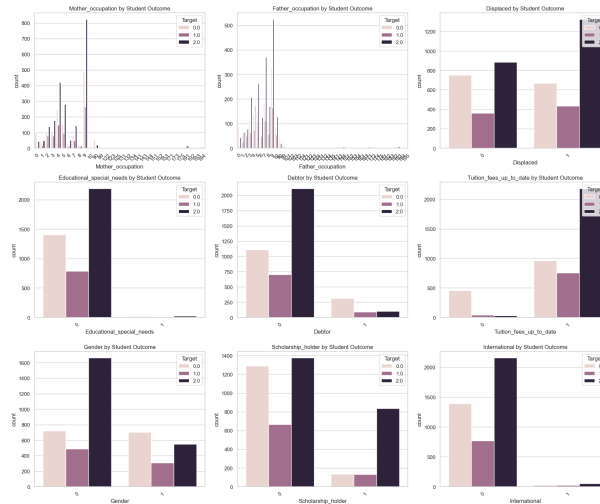


Figure 4.4: Bivariate analysis of categorical features with target features

show a drop-off in the second semester. Approved units directly measure a student’s ability to meet academic requirements and progress toward graduation. Approved credits confirm achievement, unlike enrollment or evaluation counts, which reflect participation or attempted effort. Students may register for or attempt many units, but it is the successful completion—i.e., approval.

Figure 4.6 shows strong multicollinearity among paired features from the first and second semesters. For example, the number of enrolled, evaluated, and approved curricular units shows consistently high correlation coefficients across terms ($r \geq 0.75$), indicating redundancy in measuring academic effort and engagement over time. This redundancy could distort model interpretability and inflate feature importance.

A set of averaged academic features was created by taking the mean of the corresponding first and second semester variables (e.g., avg_enrolled, avg_grade, avg_approved). Figure 4.7 was computed using these composite indicators, confirms the success of this transformation: the averaged features retain strong mutual correlations ($r \approx 0.75$), especially between average

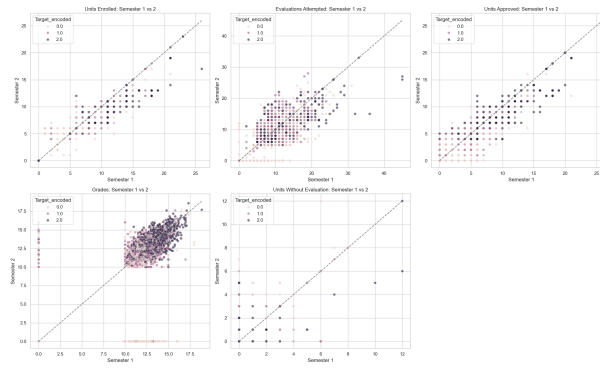


Figure 4.5: Semester progression and student outcomes

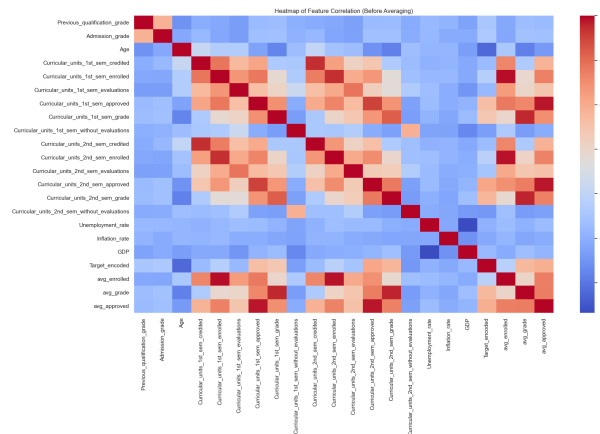


Figure 4.6: Multicollinearity of continuous features

grade and average approved units. These transformed features show correlations with the target outcome variable, with avg_grade and avg_approved showing positive coefficients of 0.55 and 0.59 respectively. This maintains their predictive relevance while reducing dimensional complexity and overfitting risk.

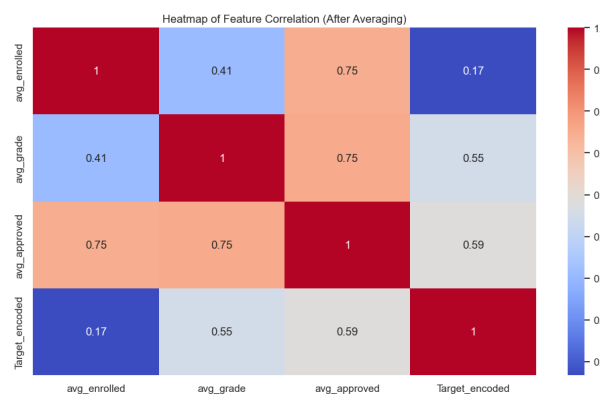


Figure 4.7: Multicollinearity of continuous features after aggregation

Figure 4.8 shows the relationship between average grades and dropout rates, reinforcing the hypothesis that academic difficulty, reflected by lower average grades, is associated with increased student attrition. For instance, Biofuel Production Technologies and Animation Design show significantly lower average grades and notably high dropout rates, suggesting these

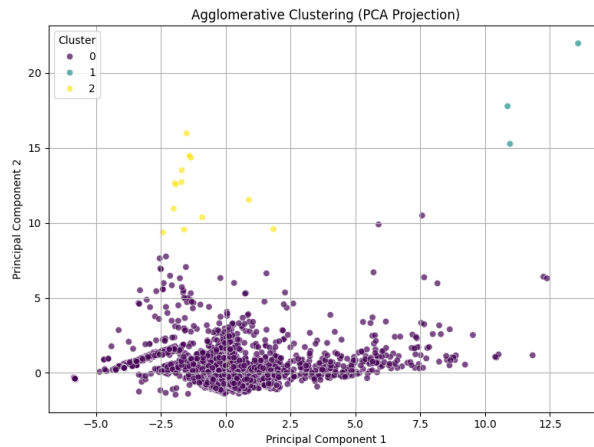


Figure 4.9: Cluster Projection using PCA

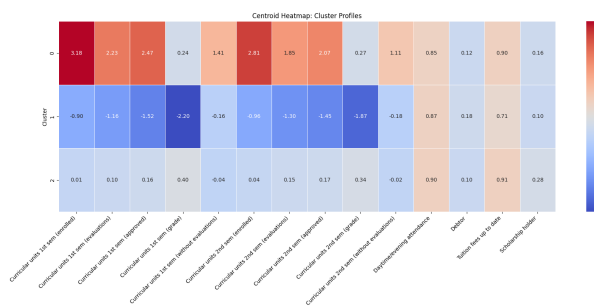


Figure 4.10: Cluster Centroids visualized using a Heatmap

2. **Cluster 1** was the highest-performing group, because the cluster showed consistently strong academic metrics and. This group showed no debt flags and full tuition payment compliance.
3. **Cluster 2** emerged as a high-risk group, typified by poor engagement, low approvals, and elevated rates of debt or missing evaluations. Their profiles aligned strongly with dropout outcomes.

SHAP analysis on a classifier trained to predict cluster membership Figure 4.11 confirmed that the most influential features were tied to academic approval and grading, especially during the first semester.

4.2.1 Statistical Validation of Cluster Differences and Outcomes

ANOVA and chi-square tests were conducted on all engagement features to confirm that these clusters were not artefacts of the clustering algorithm. All variables showed statistically significant differences between clusters ($p < 0.05$), suggesting the groupings captured meaningful and measurable differences in academic behaviour.

Although clustering was performed without using student outcome labels (Dropout, Enrolled, Graduate), post hoc analysis using crosstab analysis revealed the following as shown in Figure 4.12:

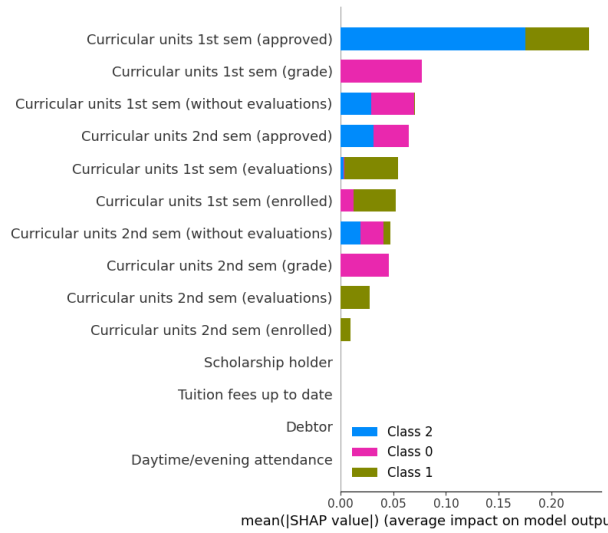


Figure 4.11: Cluster wise influential features

Table 4.1: Statistical Test Results for Feature Significance

#	Feature	Test	p-value	Significant ($p < 0.05$)
0	Curricular units 1st sem (enrolled)	ANOVA	0.0000	True
1	Curricular units 1st sem (evaluations)	ANOVA	0.0000	True
2	Curricular units 1st sem (approved)	ANOVA	0.0000	True
3	Curricular units 1st sem (grade)	ANOVA	0.0000	True
4	Curricular units 1st sem (without evaluations)	ANOVA	1.43×10^{-87}	True
5	Curricular units 2nd sem (enrolled)	ANOVA	0.0000	True
6	Curricular units 2nd sem (evaluations)	ANOVA	0.0000	True
7	Curricular units 2nd sem (approved)	ANOVA	0.0000	True
8	Curricular units 2nd sem (grade)	ANOVA	0.0000	True
9	Curricular units 2nd sem (without evaluations)	ANOVA	8.07×10^{-56}	True
10	Daytime/evening attendance	Chi-square	1.11×10^{-2}	True
11	Debtor	Chi-square	9.38×10^{-8}	True
12	Tuition fees up to date	Chi-square	5.01×10^{-48}	True
13	Scholarship holder	Chi-square	4.95×10^{-23}	True

1. Cluster 1 consisted exclusively of graduates (100%)
2. Cluster 2 had a 66.7% dropout rate, consistent with its low engagement profile
3. Cluster 0 was a mixed segment with 50% graduates and 32% dropouts.

This suggests that the behavioural profiles uncovered via clustering are highly relevant for understanding academic outcomes, even without supervised labels.

However, this finding was contrasted by the limited predictive performance of Random Forest model trained on the clusters to infer student outcomes. Figure 4.13 and classification accuracy (50%) show substantial misclassification, particularly in distinguishing dropouts from graduates. This suggests that while the clusters are behaviourally meaningful, they do not directly mirror outcome categories one-to-one. In particular, Cluster 0's mixed outcome composition, makes it harder to use the clusters alone to accurately predict student outcomes.

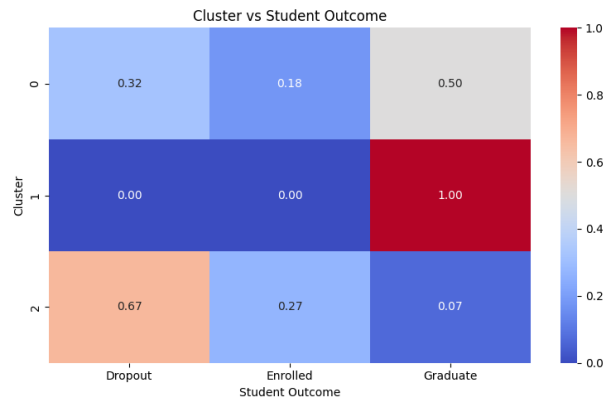


Figure 4.12: Crosstab evaluations

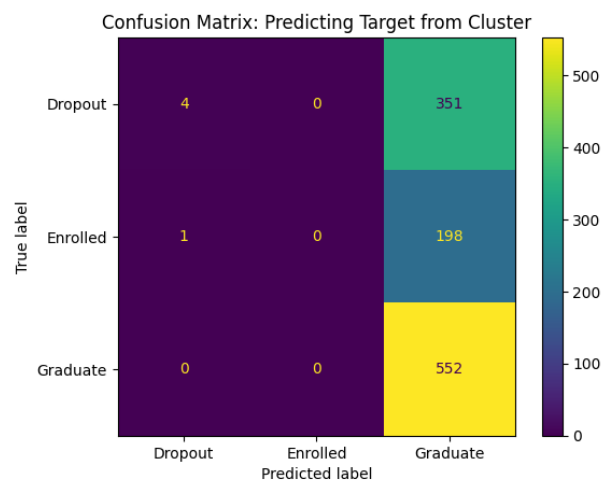


Figure 4.13: Confusion matrix of the a Random Forest model trained on the selected features and cluster labels

The clustering approach succeeded in segmenting students into behaviorally distinct groups with strong internal consistency and statistical significance. These groups aligned meaningfully, but not perfectly, with academic outcomes. The model's poor performance in predicting outcomes from clusters suggests that the behavioral groupings reflect general patterns rather than fixed results. Students with similar engagement may still end up with different outcomes due to other features unaccounted for in the clustering process. The unsupervised model is inadequate in making predictions in isolation.

4.3 Supervised Learning

4.3.1 Performance Comparison Across Multi-class Classification Models

The objective is to predict one of three possible academic outcomes for each student: Dropout, Enrolled, or Graduate.

4.3.1.1 Performance with Basic Feature Engineering

The baseline Random Forest model demonstrates modest overall performance, with a balanced accuracy of 0.692 and an F1 score of 0.701. Figure 4.14 shows strong identification of “Dropout” and “Graduate” students but a consistent failure to correctly classify “Enrolled” students, who are frequently misassigned to the other classes. This highlights the difficulty in predicting students who are still enrolled, especially when the model only has access to average academic performance across two semesters.

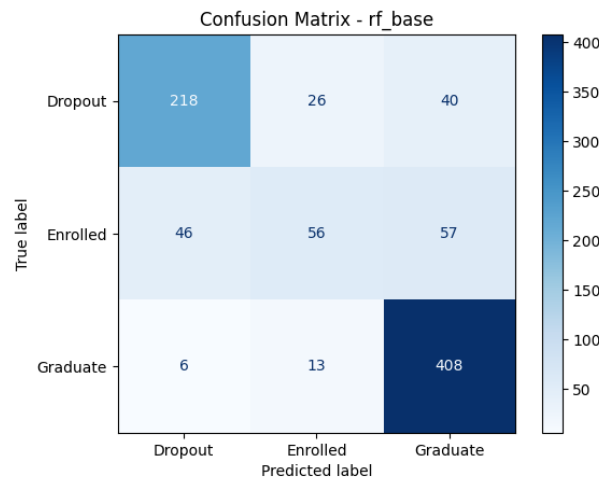


Figure 4.14: Confusion matrix of the baseline Random Forest model using basic feature engineering for multi-class student outcome prediction

Tuning the Random Forest improves its ability to balance class predictions. The tuned Random Forest increases its balanced accuracy to 0.711 and F1 score to 0.711. Figure 4.15 shows a more even allocation of predictions across classes, especially with improved detection of the “Enrolled” class. However, this leads to more mistakes in predicting “Graduate” students, showing a trade-off between better detection of enrolled students and increased errors for the other categories.

The baseline XGBoost model achieves slightly better class separation, with a balanced accuracy of 0.712 and F1 score of 0.716. Figure 4.16 shows that the model handles all three classes with relative stability, yielding fewer extreme misclassifications. The performance gains are consistent with XGBoost’s strength in handling non-linear relationships in structured data.

When tuned, XGBoost outperforms all other models on the basic feature set, achieving a balanced accuracy of 0.721, F1 score of 0.713, and AUC of 0.894. Figure 4.17 shows that the model correctly identifies the highest number of “Enrolled” students while maintaining strong performance on the other classes. This shows that tuning hyperparameters helps XGBoost make more confident distinctions in overlapping feature spaces.

The voting ensemble model, which combines the tuned Random Forest and tuned XGBoost, offers slightly lower performance in terms of balanced accuracy (0.694) and F1 score (0.701), but achieves the highest AUC (0.897) among the basic models. Figure 4.24 Its confusion matrix

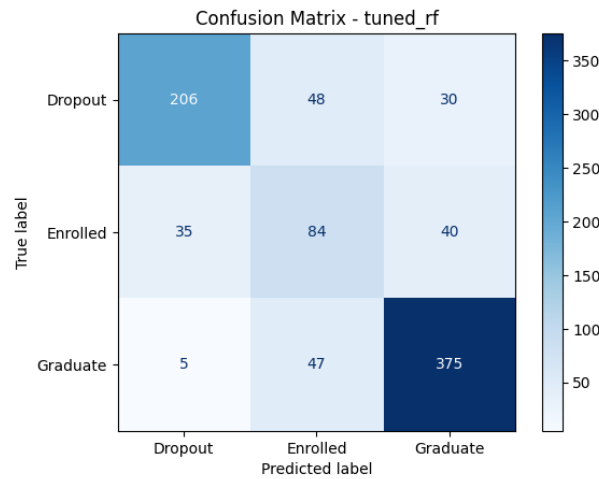


Figure 4.15: Confusion matrix of the tuned Random Forest model using basic feature engineering for multi-class student outcome prediction

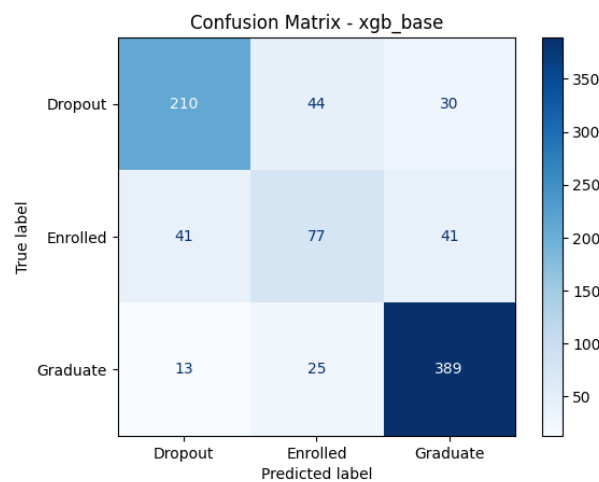


Figure 4.16: Confusion matrix of the baseline XGBoost model using basic feature engineering for multi-class student outcome prediction

shows fewer extreme errors across the classes, suggesting that the ensemble helps balance out the mistakes made by individual models.

4.3.1.2 Performance with Advanced Feature Engineering

The baseline Random Forest shows a drop in performance, with balanced accuracy falling to 0.674 and F1 to 0.682. Figure 4.19 shows the misclassification pattern remains skewed, particularly in its inability to improve predictions for “Enrolled” students despite the availability of richer behavioral, economic, and demographic features. The model is either unable to effectively use the new features or is adversely affected by increased feature dimensionality and noise.

In contrast, the tuned Random Forest shows measurable improvement, achieving a balanced accuracy of 0.708 and F1 score of 0.704. Figure 4.20 shows enhanced ability to classify

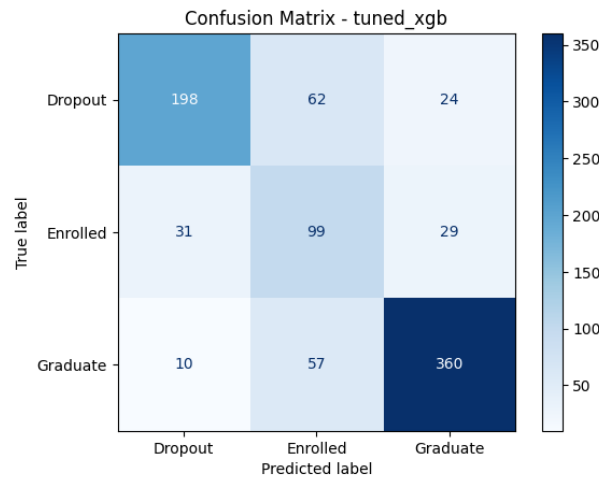


Figure 4.17: Confusion matrix of the tune XGBoost model using basic feature engineering for multi-class student outcome prediction

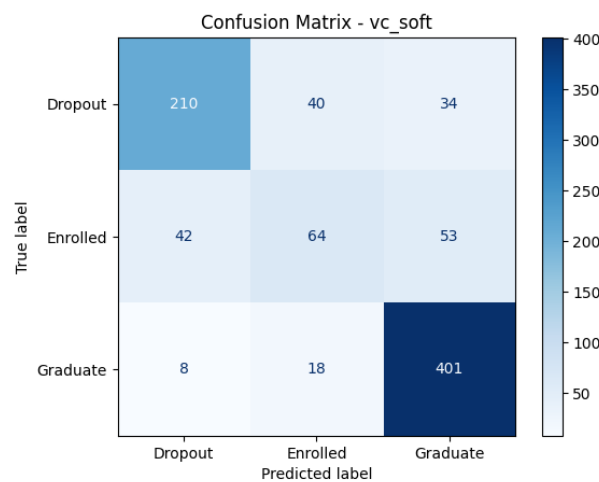


Figure 4.18: Confusion matrix of the ensemble model using basic feature engineering for multi-class student outcome prediction

“Enrolled” students, but this comes with a cost: a noticeable number of “Graduate” students are misclassified as “Enrolled,” indicating that richer features may cause the model to shift predictions toward the “Enrolled” class in ambiguous cases.

The baseline XGBoost maintains competitive performance with a balanced accuracy of 0.704 and F1 score of 0.705. Figure 4.21 shows good predictive balance across all classes and fewer critical misclassifications.

The tuned XGBoost again stands out, matching its performance on the basic set (balanced accuracy of 0.712, F1 score of 0.695, AUC of 0.886). However, Figure 4.22 shows that the model suffers from increased overlap between “Enrolled” and both “Dropout” and “Graduate” classes. The model appears to use engineered features to reassign borderline cases to the middle class, improving recall for “Enrolled” but weakening class precision.

The ensemble model achieves a balanced accuracy of 0.705 and F1 score of 0.709, with an AUC of 0.890. Figure 4.23 indicates reduced extremity in misclassification and more uniform

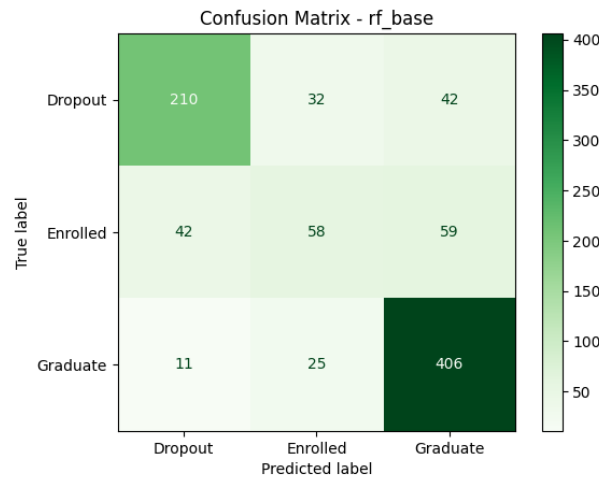


Figure 4.19: Confusion matrix of the baseline Random Forest model using advanced feature engineering for multi-class student outcome prediction

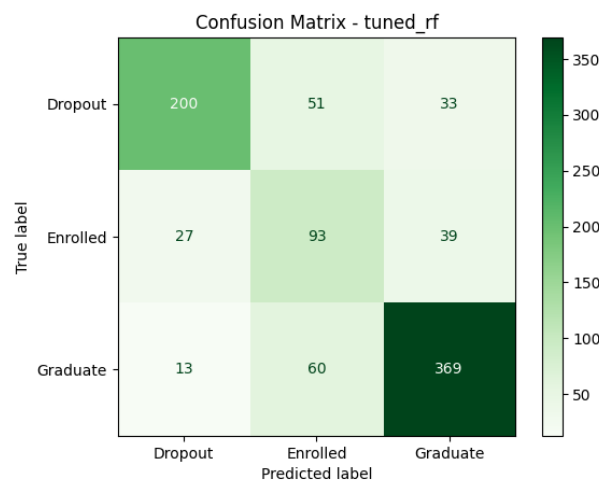


Figure 4.20: Confusion matrix of the tuned Random Forest model using advanced feature engineering for multi-class student outcome prediction

treatment of all classes, making it suitable for applications where model stability is prioritized over peak accuracy.

Across both basic and advanced feature sets, the tuned XGBoost model consistently delivers the strongest performance, especially in identifying students who are still enrolled — a class that remains the most difficult to classify correctly. In multiclass classification tasks like this, predicting transitional states such as “Enrolled” is challenging because these students often share characteristics with both dropouts and graduates, making them harder to distinguish. Ensemble models show more stable and balanced behavior overall. While they help reduce extreme misclassifications, they do not always outperform the tuned XGBoost in terms of accuracy or F1 score. Importantly, using advanced features does not always lead to better results. In several models, it actually increases confusion between “Graduate” and “Enrolled” students. This suggests that while additional features can be useful, they must be carefully selected and matched with models that are capable of handling complex, high-dimensional

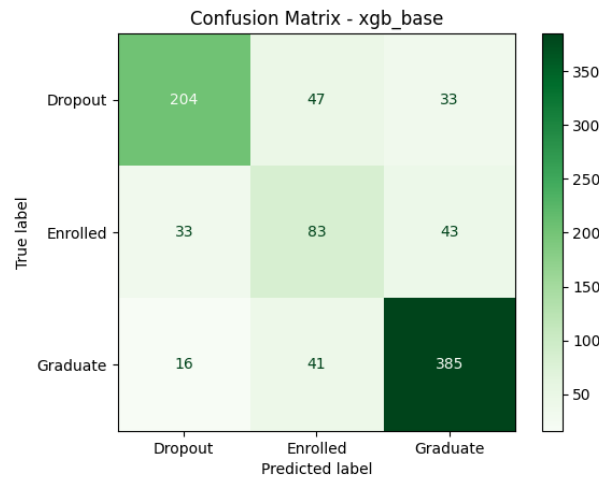


Figure 4.21: Confusion matrix of the baseline XGBoost model using advanced feature engineering for multi-class student outcome prediction

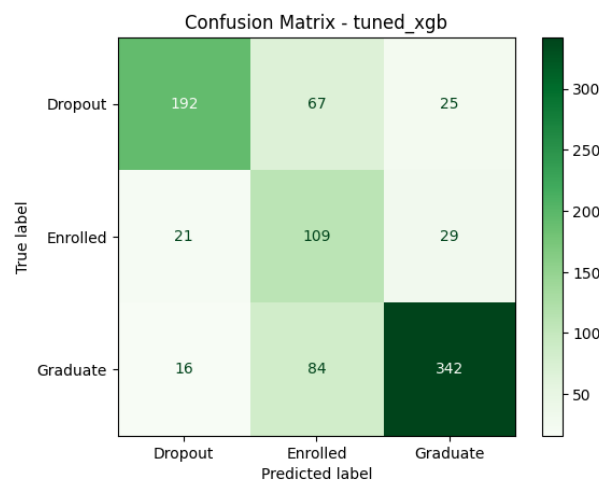


Figure 4.22: Confusion matrix of the tuned XGBoost model using advanced feature engineering for multi-class student outcome prediction

data without becoming less accurate or more biased.

4.3.2 Performance Comparison Across Binary-class Classification Models

The objective is distinguishing between two terminal academic outcomes: Dropout and Graduate. The “Enrolled” class was excluded to improve class separability and focus on identifying students at the most significant risk of discontinuation.

4.3.2.1 Performance with Basic Feature Engineering

The baseline Random Forest model achieved a balanced accuracy of 0.906 and an F1 score of 0.914. Figure 4.25 reveals relatively symmetrical performance across both classes, with 1,191 correct dropout and 2,077 correct graduate predictions. Misclassifications are present but limited, showing that the model is robust even without tuning.

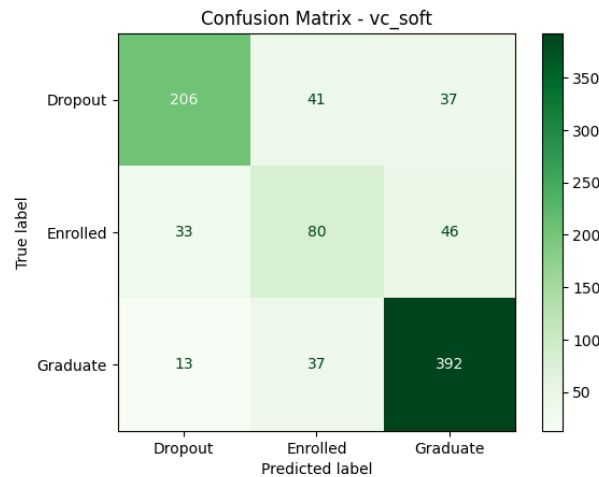


Figure 4.23: Confusion matrix of the ensemble model using basic feature engineering for multi-class student outcome prediction

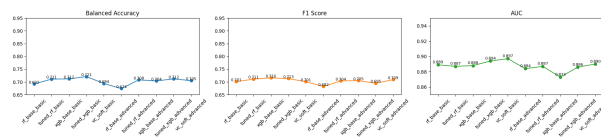


Figure 4.24: Multi-class model comparison

The tuned Random Forest model exhibits similar performance, with balanced accuracy and F1 scores of 0.903 and 0.910, respectively. Figure 4.26 shows while it reduces errors for “Dropout” predictions slightly (increasing correct predictions from 1,191 to 1,195), it slightly increases the number of misclassified “Graduate” instances. This shows a trade-off where optimisation leads to a redistribution of errors rather than a net gain in overall performance.

The baseline XGBoost model also performs well, matching the tuned Random Forest with a balanced accuracy of 0.903 and an F1 score of 0.910. Figure 4.27 shows slightly more conservative predictions favouring the “Dropout” class. This shows that XGBoost handles class imbalance effectively when paired with sample weighting, even without tuning.

The tuned XGBoost model stands out as the top-performing individual classifier, with a balanced accuracy of 0.915, an F1 score of 0.921, and an AUC of 0.957. It shows high precision and recall across both classes, with 1,226 correctly predicted dropouts and 2,065 correctly predicted graduates. Misclassification rates are the lowest among all individual models as shown in Figure 4.28.

The ensemble voting classifier, which combines the tuned Random Forest and XGBoost using soft voting, performs nearly as well. It achieves a balanced accuracy of 0.912, an F1 score of 0.919, and the highest AUC of 0.958 among all models. Figure 4.29 confirms stable and reliable predictions, with 1,209 “Dropout” and 2,075 “Graduate” cases correctly classified. While it does not surpass tuned XGBoost in raw accuracy or F1 score, its high AUC and consistent classification across folds suggest that the ensemble is particularly effective in reducing variance and maximising decision confidence.

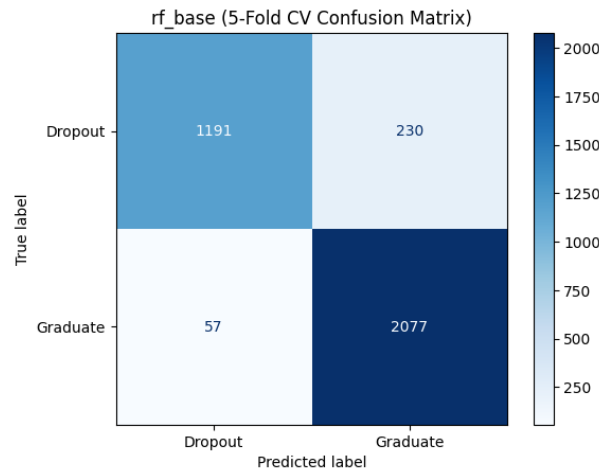


Figure 4.25: Confusion matrix of the baseline Random Forest model using basic feature engineering for binary class student outcome prediction

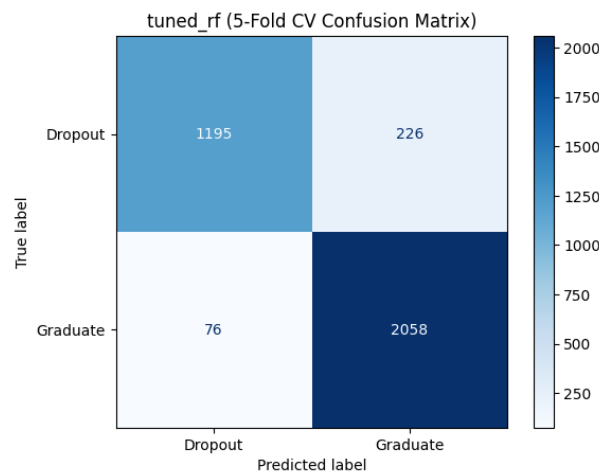


Figure 4.26: Confusion matrix of the tuned Random Forest model using basic feature engineering for binary class student outcome prediction

4.3.2.2 Performance with Advanced Feature Engineering

The baseline Random Forest achieved a balanced accuracy of 0.895, an F1 score of 0.902, and an AUC of 0.949. Figure 4.30 shows relatively balanced classification, with 1,189 correctly predicted “Dropout” cases and 2,107 correctly identified “Graduate” cases. However, it also shows a moderate level of misclassification, with 232 “Dropout” students labelled as graduates, indicating some sensitivity to the added feature interactions.

The tuned Random Forest slightly improved upon the baseline, reaching a balanced accuracy of 0.898 and an F1 score of 0.903. Although precision increased for the “Dropout” class (1,209 correct predictions), there was a slight uptick in misclassified “Graduate” students (119 labelled as dropouts) as shown in figure 4.31. This suggests a shift in the decision boundary influenced by improved handling of behavioural and parental features, though not enough to dramatically improve overall separation.

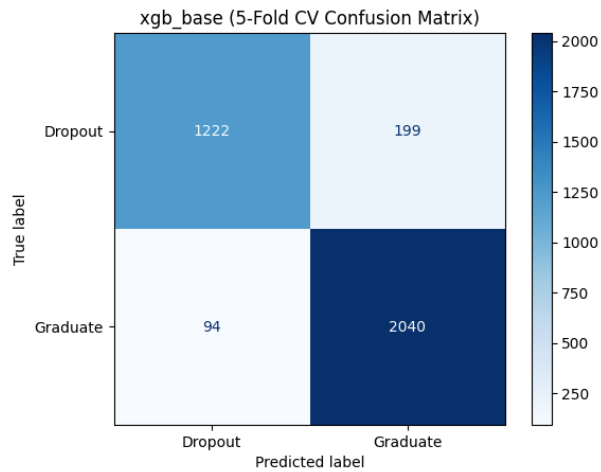


Figure 4.27: Confusion matrix of the baseline XGBoost model using basic feature engineering for binary class student outcome prediction

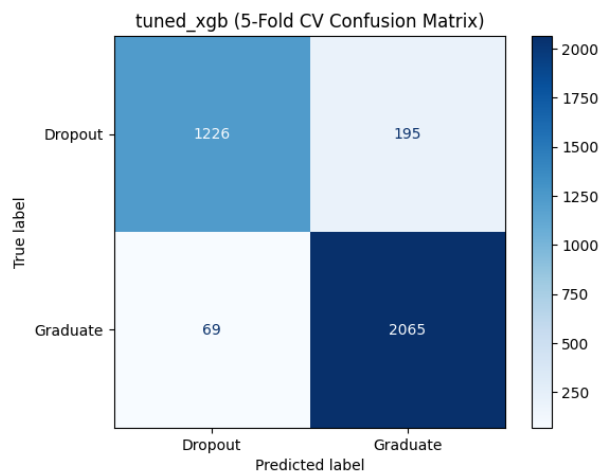


Figure 4.28: Confusion matrix of the tuned XGBoost model using basic feature engineering for binary class student outcome prediction

The baseline XGBoost underperformed and achieved a balanced accuracy of 0.889 and an F1 score of 0.895. Figure 4.32 indicates a slightly more conservative pattern, with 1,187 correct “Dropout” predictions and 2,084 “Graduate” predictions.

The tuned XGBoost emerged as the best individual performer regarding overall balance and class-wise prediction. It achieved a balanced accuracy of 0.902, F1 score of 0.907, and AUC of 0.952. It correctly classified 1,215 dropouts and 2,096 graduates, while reducing false positives compared to its baseline version showing in figure 4.33.

The voting ensemble classifier demonstrated the most stable performance profile. With a balanced accuracy of 0.898, F1 score of 0.904, and the highest AUC of 0.953, the ensemble avoided extreme misclassifications and provided a well-balanced prediction across both classes. Figure 4.34 shows 1,191 correct “Dropout” and 2,114 correct “Graduate” predictions, indicating minimal performance trade-offs. Although it does not outperform the tuned XGBoost in all metrics, its marginally higher AUC suggests that its predictive confidence is better calibrated.

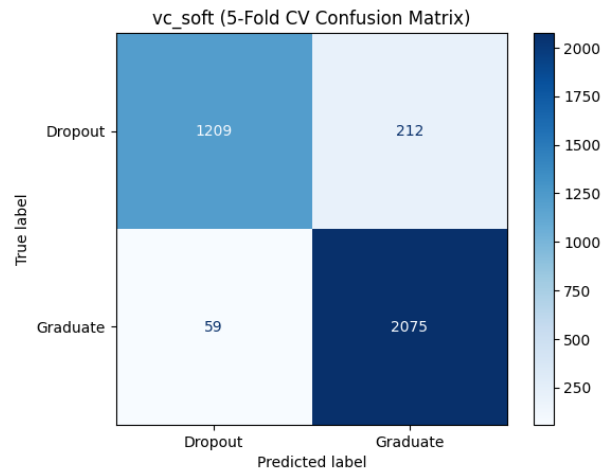


Figure 4.29: Confusion matrix of the tuned XGBoost model using basic feature engineering for binary class student outcome prediction

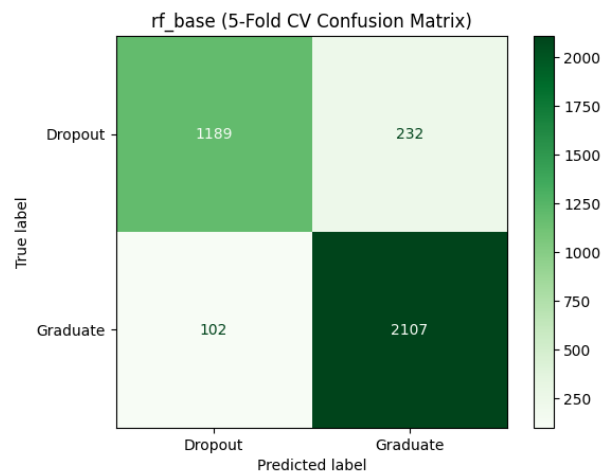


Figure 4.30: Confusion matrix of the baseline Random Forest model using advanced feature engineering for binary class student outcome prediction

The binary framing of the classification task allows all models to perform more effectively than in the multiclass scenario. Without the transitional “Enrolled” class, the models face less ambiguity in decision boundaries, enabling clearer separation between at-risk and successful students. The application of advanced features yielded consistently strong results across all five models, though the performance improvements over the basic feature set were generally marginal and sometimes mixed. This suggests that richer feature representations offer added informational value but introduce added complexity that not all models handle equally well.

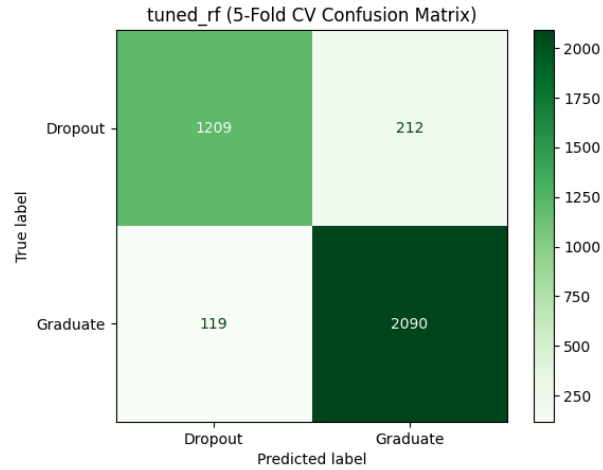


Figure 4.31: Confusion matrix of the tuned Random Forest model using advanced feature engineering for binary class student outcome prediction

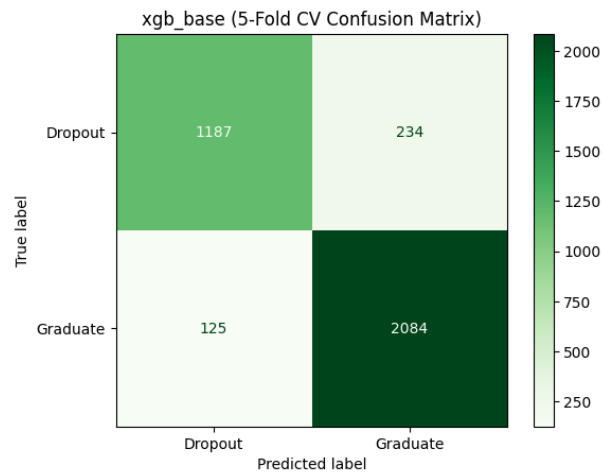


Figure 4.32: Confusion matrix of the baseline XGBoost model using advanced feature engineering for binary class student outcome prediction

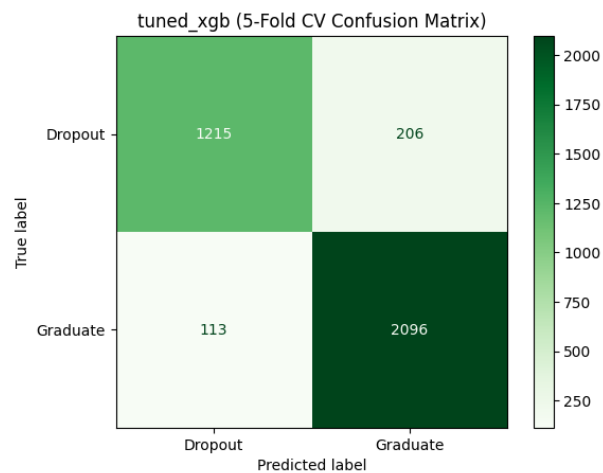


Figure 4.33: Confusion matrix of the tuned XGBoost model using advanced feature engineering for binary class student outcome prediction

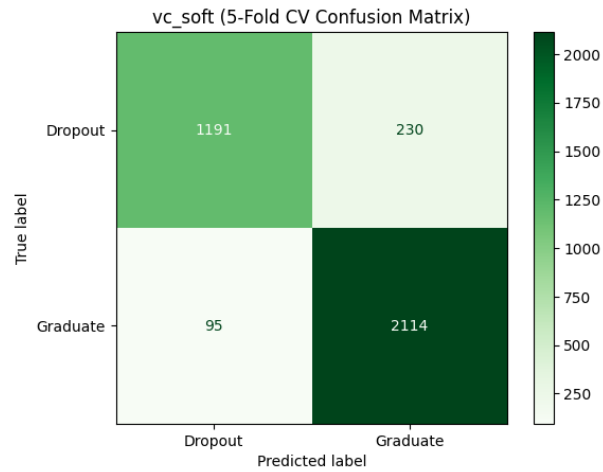


Figure 4.34: Confusion matrix of the tuned XGBoost model using advanced feature engineering for binary class student outcome prediction

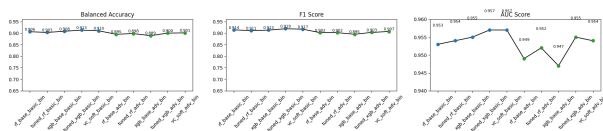


Figure 4.35: Binary class model comparison

Chapter 5

Discussion

5.1 Comparison with existing studies

This research explored multiclass and binary classification to predict student outcomes, revealing a clear performance gap. Models consistently performed better on the binary task (Dropout vs. Graduate), with tuned XGBoost and ensemble classifiers achieving balanced accuracy above 0.91 and AUC near 0.96. These results mirror the findings of Hassan et al. (2024) and Villar and de Andrade (2024), where binary dropout prediction yielded higher recall and precision due to more precise class boundaries. In contrast, multiclass classification introduced greater ambiguity, particularly for the "Enrolled" class, leading to more misclassifications and lower F1 scores. This challenge is also noted in Alhazmi and Sheneamer (2023), where transitional academic states were harder to classify due to feature overlap with both terminal outcomes.

The findings from Realinho et al. (2022), which have contributed to creating the dataset, further reinforce the structure and challenges of this study's three-class classification task. Their use of the same outcome classes and their identification of significant class imbalance, particularly the underrepresentation of the "Enrolled" class, closely mirror the difficulties encountered in this research. Their work also validates the predictive strength of academic performance variables, especially approvals and grades in the second semester, which aligns with the SHAP and permutation importance results observed here. While they explored multiple ensemble tree models, this study extends that work by evaluating a soft voting ensemble, which showed superior AUC in both binary and multiclass tasks. This suggests that while their models performed well, integrating tuned base models through ensembling offers a more stable solution for imbalanced and structurally complex academic datasets.

Feature engineering had mixed effects. Averaged academic metrics across semesters offered a strong baseline performance, aligning with studies like Arora (2023) that found early academic indicators to be dominant predictors. However, incorporating socioeconomic and behavioural features did not consistently improve results—in some models, it introduced noise and reduced precision. This diverges from studies like Anwar (2025) and N. et al. (2020),

where advanced features and hybrid modelling improved prediction. Across all experiments, tree-based models—especially tuned XGBoost—remained top performers, reflecting a typical pattern in the literature. The ensemble soft-voting classifier offered the best AUC in multi-class and binary tasks, similar to findings in Kim et al. (2023), where combining CatBoost and XGBoost improved overall stability. While ensemble models did not consistently outperform tuned XGBoost in raw accuracy, their smoother error distribution suggests value in applications requiring consistent, lower-variance predictions.

The clustering approach in this study aligns closely with previous research that used unsupervised methods to identify behavioural patterns linked to dropout. Like Palani et al. (2021) and Iam-On and Boongoen (2017), this research found clear clusters with statistically distinct academic engagement and administrative profiles. The strong silhouette score, high variance between clusters, and actual outcomes (e.g., 100% graduates in Cluster 1, 66.7% dropouts in Cluster 2) echo prior findings that unsupervised learning can reveal meaningful risk segments. SHAP analysis confirmed that early academic metrics (especially first-semester approvals and grades) were key differentiators, reinforcing the literature’s emphasis on early performance as a dropout predictor.

However, unlike hybrid models used in Anwar (2025) and N. et al. (2020), this study’s clusters were ineffective for direct outcome prediction. A simple Random Forest classifier trained on clusters misclassified many students, especially from the mid-range group (Cluster 0). It shows that unsupervised groupings can be helpful in behavioural segmentation but do not map neatly onto outcome categories. This contrasts with studies that used clustering as a preprocessing step or generated pseudo-labels to boost prediction. This research did not incorporate time-based engagement features for clustering or cluster-specific model tuning, which have improved model performance in prior studies. Clustering adds interpretability and supports risk profiling, but it cannot be used alone for predictive modelling. However, these cluster findings can be used as additional derived features to enhance prediction.

5.2 Research questions

RQ1: What patterns in student academic and financial records are associated with different outcomes — dropout, continued enrollment, or graduation?

The research proves that academic performance, particularly curricular unit approvals and grades in the first and second semesters, is strongly associated with student outcomes. These findings are consistent across unsupervised clusters, SHAP analysis, and correlation matrices. Additionally, administrative variables such as tuition fee status, scholarship receipt, and debtor status also showed statistical significance and partial alignment with outcomes. However, the patterns for the “Enrolled” class were less distinct. This group frequently overlapped with dropout and graduate classes in feature space, leading to high misclassification rates. This suggests that available static features from the first year may be insufficient to capture the

nuance of continued enrollment fully. Future work could incorporate temporal academic data, attendance patterns, and engagement logs to better address this question to track academic trajectory more precisely over time.

RQ2: Can students be grouped based on shared academic behaviors and engagement, and do these groups show different levels of dropout risk?

The clustering results demonstrate that students can be meaningfully grouped based on academic behaviour and financial engagement, with statistically significant distinctions across engagement features. Post hoc analysis shows that these clusters do correlate with different dropout risks, confirming the theoretical expectation. For instance, Cluster 2—with the lowest academic engagement—had a 66.7% dropout rate, while Cluster 1 consisted entirely of graduates. However, the predictive utility of these clusters was limited. When cluster labels were used to infer outcomes through classification, the resulting model struggled to distinguish among actual outcomes, particularly for students in Cluster 0, which had a mixed profile. This highlights a limitation: untuned unsupervised clusters can offer insight into student behaviours but are not directly predictive of outcomes without further contextual data or model layering. This question was addressed meaningfully but not comprehensively, and would benefit from hybrid modelling strategies that integrate clustering with downstream supervised learning or refine clusters with feature engineering.

RQ3: To what extent can student dropout be predicted using academic, financial, and enrollment-related variables available during the first year of study?

This research shows that dropout can be predicted with high accuracy using only first-year data, especially in the binary classification setting. Tuned XGBoost achieved a balanced accuracy of 0.915 and an AUC of 0.957, indicating strong model performance. This suggests that early academic records and financial status offer a reliable basis for identifying at-risk students, aligning with findings in the literature such as \citet{villar2024supervised} and \citet{app14177593}. However, performance declined in the multiclass scenario, especially for "Enrolled" students. This implies that first-year data alone is insufficient for accurate multiclass prediction, particularly for intermediate outcomes. Thus, while RQ3 is well answered for dropout vs. graduate predictions, it remains only partially answered for more nuanced classification tasks.

RQ4: Which features most strongly contribute to the likelihood of student dropout, and how can this information help in formulating strategies for intervention?

The feature importance results from SHAP, and correlation tests provide strong evidence that academic approvals, grades, and tuition payment status are the most impactful predictors of dropout. These findings are consistent with past studies and give institutions actionable variables for early intervention—students with low first-semester approvals and unpaid fees can be prioritised for academic and financial support programs. However, the interpretability of more complex models like XGBoost still requires cautious communication, particularly when translating feature importance into student-level decisions. Additionally, while features showed

statistical relevance, their intervention value remains context-dependent; for example, being a scholarship holder may reduce dropout risk in some settings but not others.

Chapter 6

Conclusion

6.1 Limitations

There are inherent limitations in the modelling choices and analytical strategies that should be acknowledged.

Though effective in identifying behaviorally distinct groups, the unsupervised clustering approach showed limited predictive utility when used as input for classification models. The resulting clusters, especially Cluster 0 with mixed engagement levels, were behaviorally meaningful but did not translate well into outcome prediction. This step remains primarily descriptive rather than predictive in the research methodology.

While two classification models were explored, the analysis was constrained to ensemble tree-based models and a basic voting ensemble. Although these models are powerful and interpretable, they may not fully capture the data's complex non-linear interactions. The ensemble voting approach combined predictions but did not dynamically learn from model strengths and weaknesses, limiting the advantages of ensembling. More sophisticated techniques like stacking or boosting with customised loss functions could improve performance.

The research relied on feature averaging and static aggregations (e.g., average grade across semesters) for feature engineering, which may have diluted academic progression or recovery information. This flattened representation of student behaviour may have contributed to difficulties distinguishing the "Enrolled" class. It may have masked meaningful trends that could be captured through temporal or sequential modelling. The new features calculated in the advanced feature engineering should have been tested for feature importance before training models to reduce noise generated from the calculated features.

Although SHAP analysis and permutation feature importance supported model interpretability, these methods were applied post hoc and on the cluster labels. Interpretability was not embedded into the supervised model training process, which limits the ability to impose constraints or incorporate domain knowledge directly into decision-making.

These limitations do not undermine the validity of the findings, but rather define the

methodological boundaries within which the results should be interpreted.

6.2 Future Work

Several methodological enhancements could improve performance and generalizability.

Future work can explore a broader range of classification algorithms that may better handle class imbalance and complex decision boundaries. Models such as Gradient Boosted Decision Trees with monotonic constraints, Support Vector Machines with nonlinear kernels, or Bayesian classifiers could be evaluated for their ability to better capture nuanced class distinctions—particularly for the “Enrolled” group, which was challenging to classify. Additionally, deep learning architectures like fully connected neural networks or temporal models such as Recurrent Neural Networks (RNNs) could be applied to capture non-linear patterns in academic progression across semesters. Although deep learning models lack interpretability and are often treated as black boxes, they could be valuable for establishing upper-bound baseline performance against which interpretable models can be benchmarked.

Further refinements in feature engineering could enhance predictive accuracy. Instead of relying on average aggregation of semester-level metrics, modeling performance differences between semesters, or introducing rate-based indicators (e.g., approval-to-enrollment ratio), could better reflect academic trajectory. Additionally, soft clustering techniques—such as Gaussian Mixture Models or Fuzzy C-Means—could replace hard clustering to allow for probabilistic group membership, better accounting for overlapping behaviors. This may yield more flexible and informative segments, particularly for multiple groups. The cluster labels obtained from unsupervised learning could also be used as a feature to train supervised models.

Current models could be improved through regularized feature selection techniques, such as Recursive Feature Elimination (RFE), to reduce feature redundancy and noise. Similarly, using cost-sensitive learning could make models more robust for real-world use, especially in underrepresented classes. Incorporating model stacking could allow different algorithms to specialize in different parts of the feature space to improve generalization. These improvements could maximize the value of the existing dataset to achieve stronger performance while balancing the trade-offs between accuracy, interpretability in educational contexts.

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